

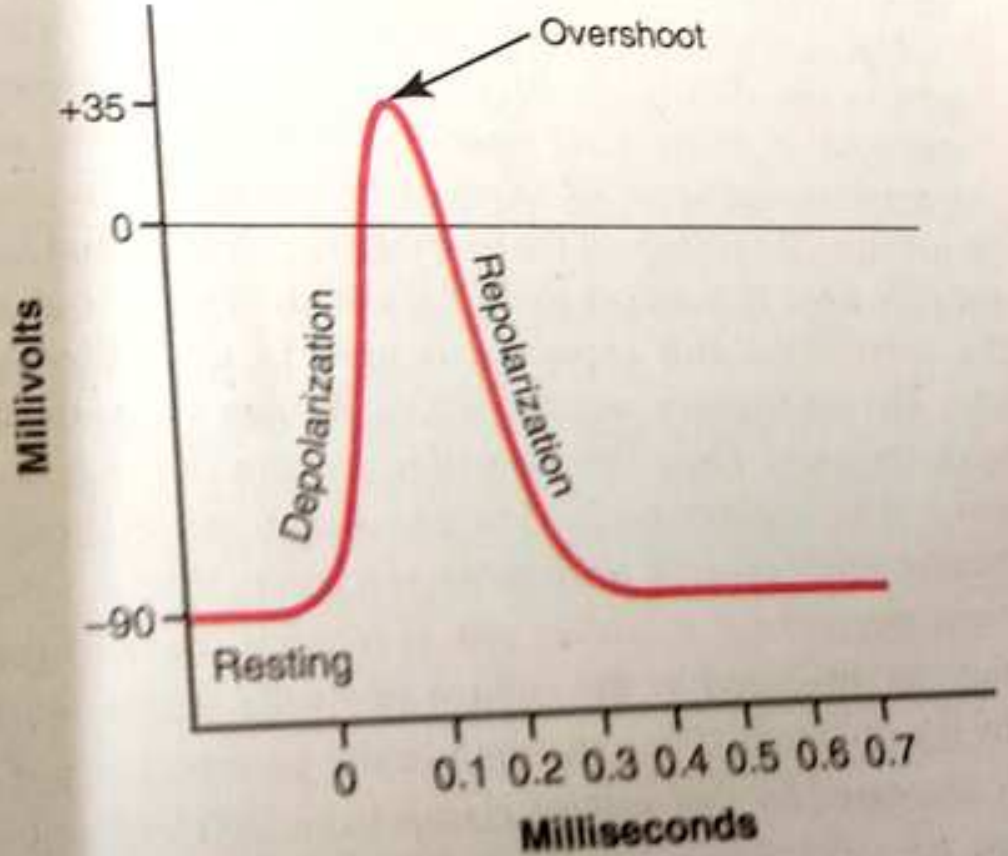


Figure 5-6. Typical action potential recorded by the method shown in the upper panel of the figure.

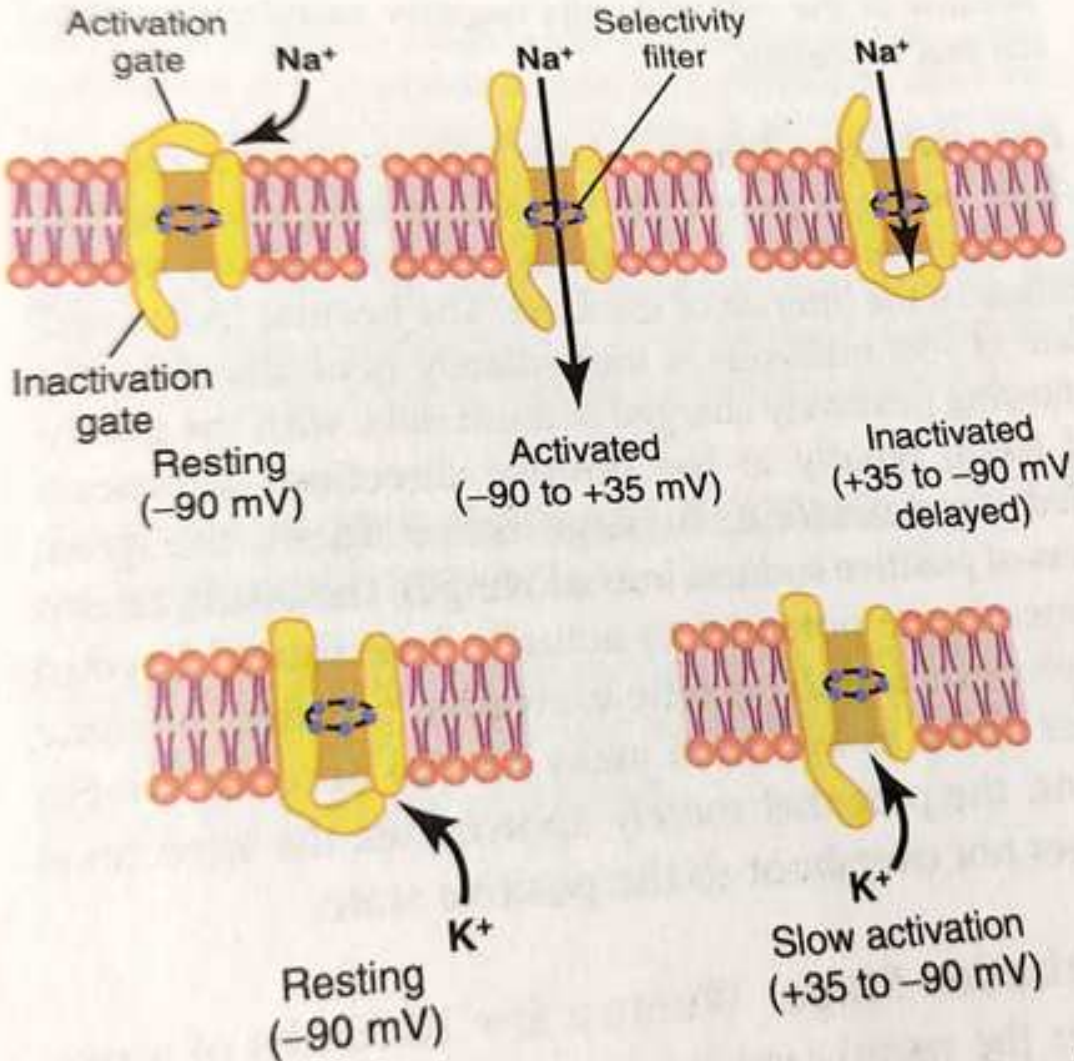
POTASSIUM CHANNELS

The necessary actor in causing both depolarization and repolarization of the nerve membrane during the action potential is the *voltage-gated sodium channel*. A *voltage-gated potassium channel* also plays an important role in increasing the rapidity of repolarization of the membrane. *These two voltage-gated channels are in addition to the* Na^+-K^+ *pump and the* K^+ *leak channels.*

Activation and Inactivation of the Voltage-Gated Sodium Channel

The upper panel of **Figure 5-7** shows the voltage-gate sodium channel in three separate states. This channel two *gates*—one near the outside of the channel called *activation gate*, and another near the inside calle *inactivation gate*. The upper left of the figure d the state of these two gates in the normal resting brane when the membrane potential is –90 milli this state, the activation gate is closed, which pre entry of sodium ions to the interior of the fibe these sodium channels.

Activation of the Sodium Channel. Wh brane potential becomes less negative th



5-7 Characteristics of the voltage-gated sodium (top) and ... successive activation and

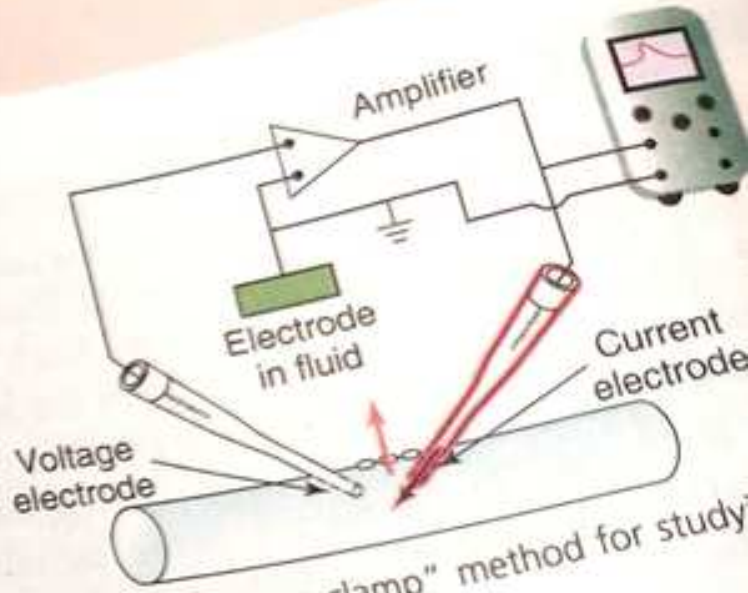


Figure 5-8. The "voltage clamp" method for study... through specific channels.

Voltage-Gated Potassium Chann... and Its Activation

The lower panel of **Figure 5-7** shows ... potassium channel in two states: duri... (left) and toward the end of the acti... During the resting state, the gate of th... is closed and potassium ions are pr... through this channel to the exterio... potential rises from –90 milliv...

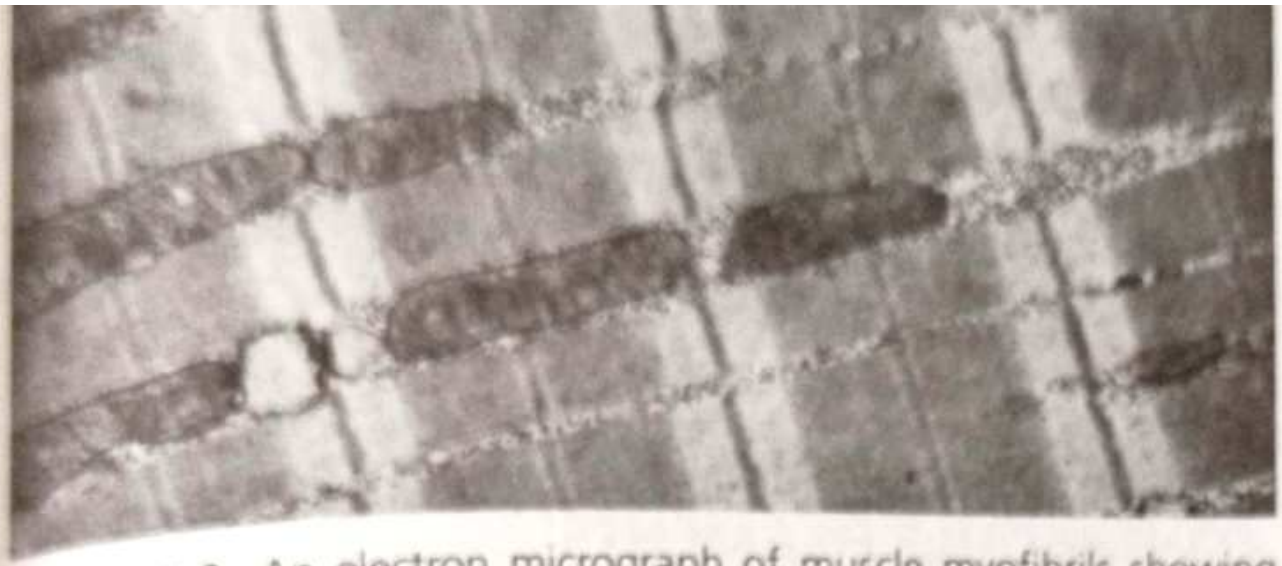


Figure 6-2. An electron micrograph of muscle myofibrils showing the detailed organization of actin and myosin filaments. Note the mitochondria lying between the myofibrils. *(From Fawcett DW: The Cell. Philadelphia: WB Saunders, 1981.)*

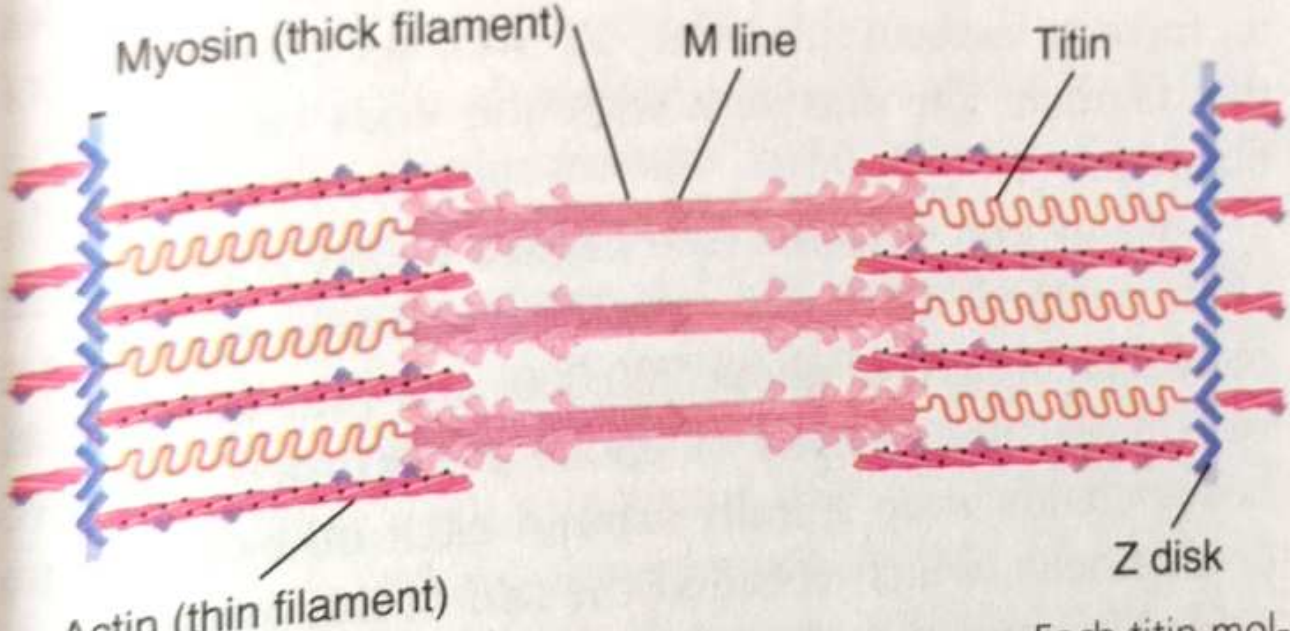


Figure 6-3. Organization of proteins in a sarcomere. Each titin molecule extends from the *Z disk* to the *M line*. Part of the titin molecule is closely associated with the myosin thick filament, whereas the rest of the molecule is springy and changes length as the sarcomere contracts and relaxes.

These springy titin mole-

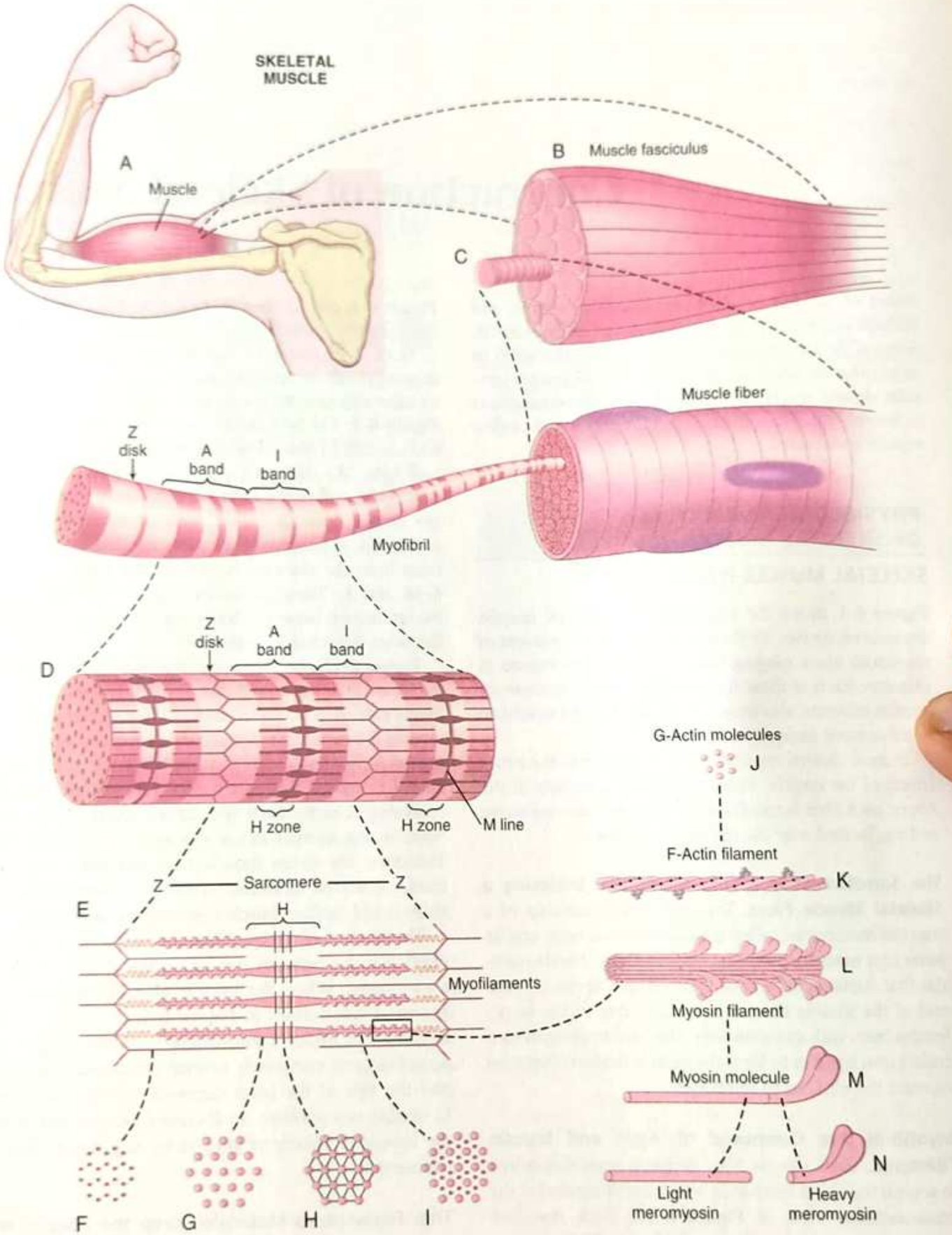
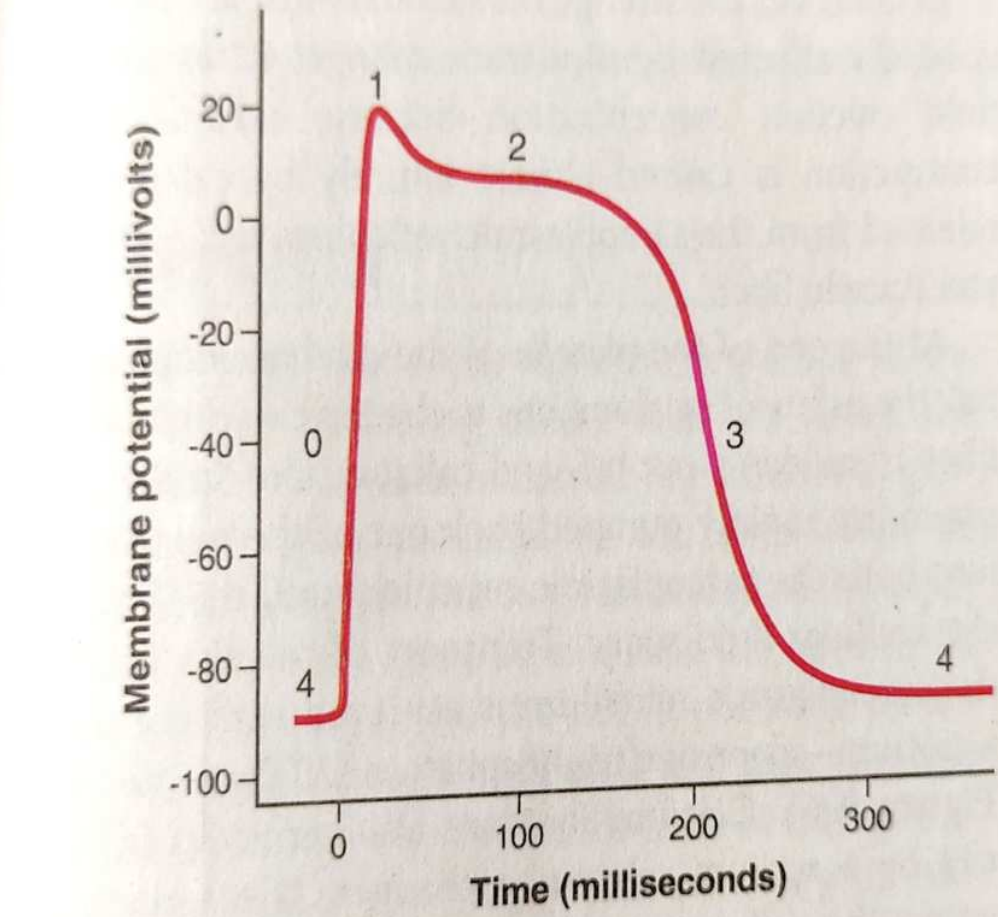


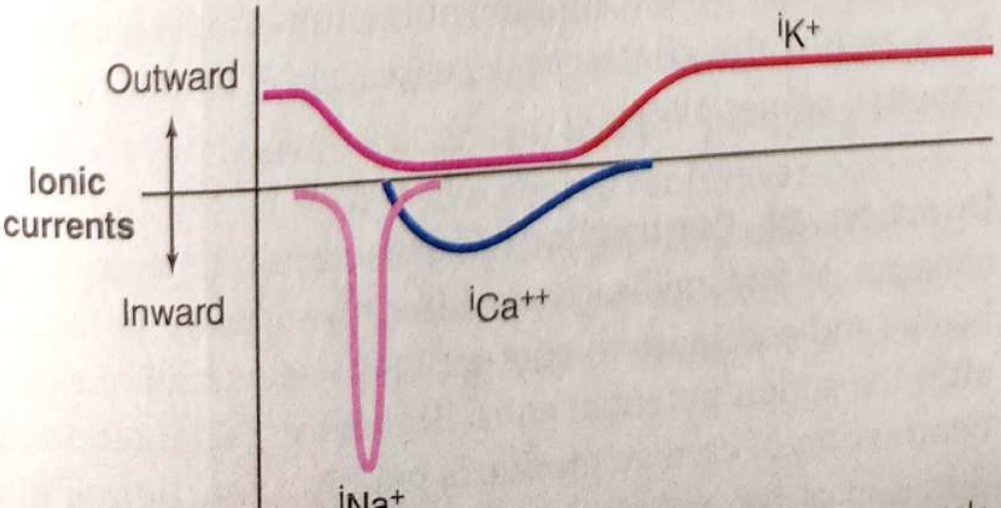
Figure 6-1. Organization of skeletal muscle, from the gross to the molecular level. *F, G, H,* and *I* are cross sections at the levels indicated.

calcium and sodium ions ceases, the membrane permeability for potassium ions also increases rapidly; this rapid loss of potassium from the fiber immediately returns the membrane potential to its resting level, thus ending the action potential.

Summary of Phases of Cardiac Muscle Action Potential. Figure 9-4 summarizes the phases of the action potential in cardiac muscle and the ion flows that occur during each phase.

Phase 0 (depolarization), fast sodium channels open. When the cardiac cell is stimulated and depolarizes, the





membrane potential becomes more positive. Voltage-gated sodium channels (fast sodium channels) open and permit sodium to rapidly flow into the cell and depolarize it. The membrane potential reaches about +20 millivolts before the sodium channels close.

Phase 1 (initial repolarization), fast sodium channels close. The sodium channels close, the cell begins to repolarize, and potassium ions leave the cell through open potassium channels.

Phase 2 (plateau), calcium channels open and fast potassium channels close. A brief initial repolarization occurs and the action potential then plateaus as a result of (1) increased calcium ion permeability and (2) decreased potassium ion permeability. The voltage-gated calcium ion channels open slowly during phases 1 and 0, and calcium enters the cell. Potassium channels then close, and the combination of decreased potassium ion efflux and increased calcium ion influx causes the action potential to plateau.

Phase 3 (rapid repolarization), calcium channels close and slow potassium channels open. The closure of calcium ion channels and increased potassium ion permeability, permitting potassium ions to rapidly exit the cell, ends the plateau and returns the cell membrane potential to its resting level.

Phase 4 (resting membrane potential) averages about –90 millivolts.

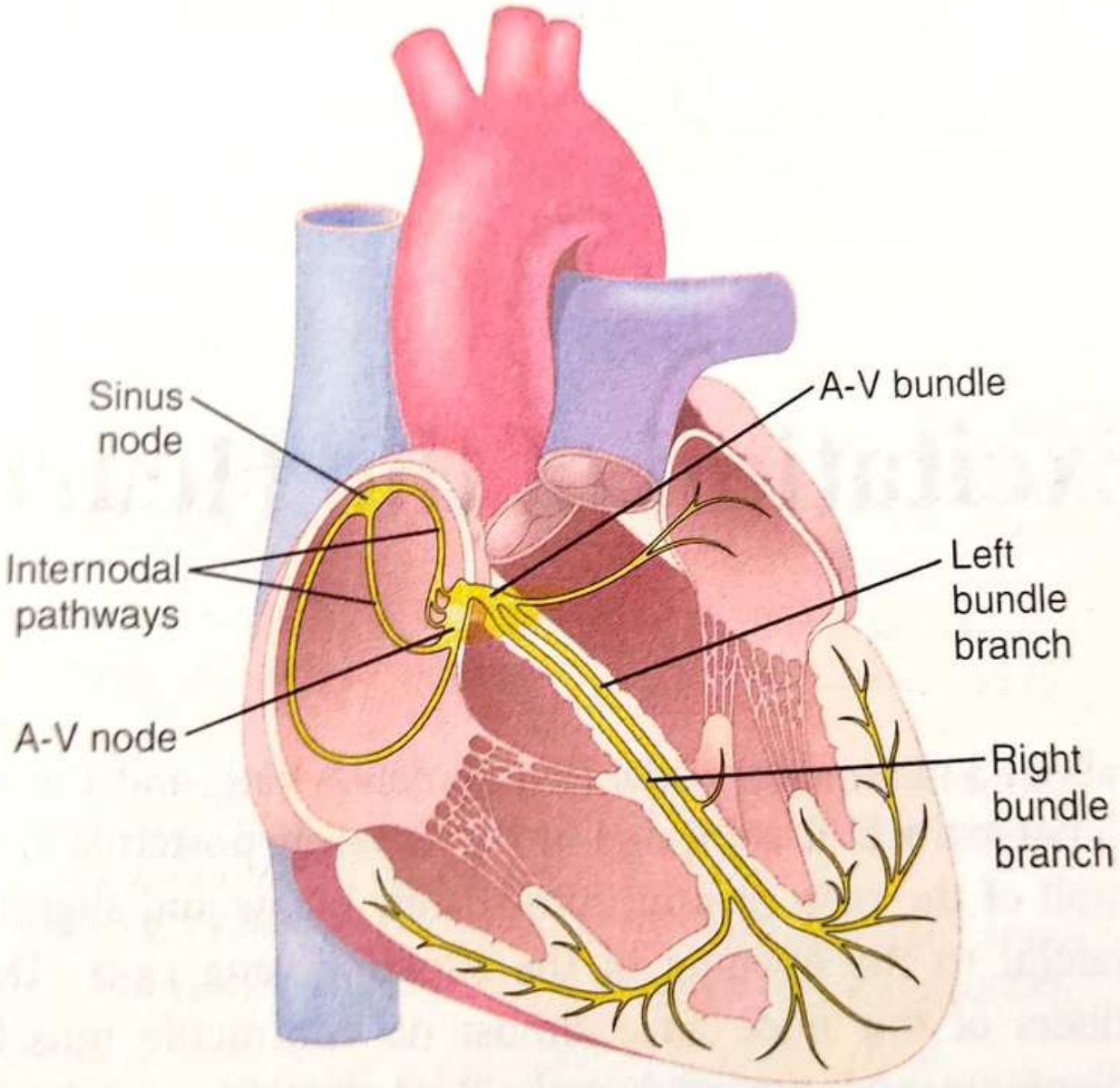
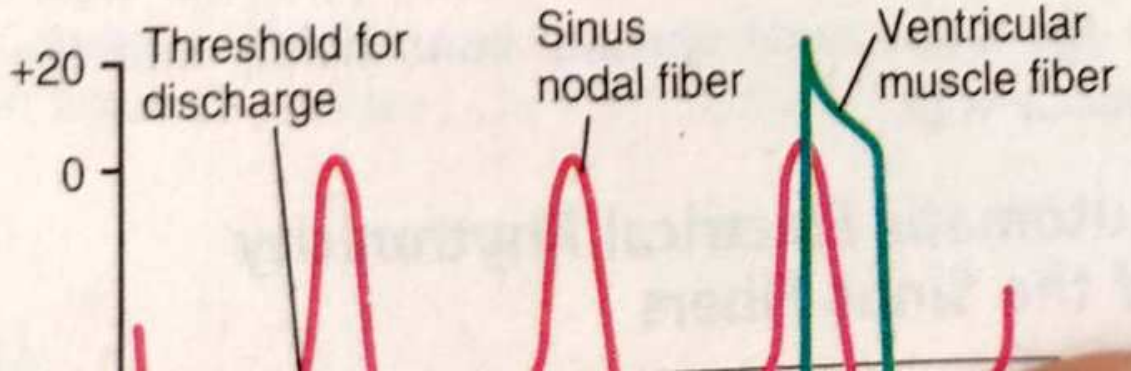


Figure 10-1. Sinus node and the Purkinje system of the heart, showing also the atrioventricular *(A-V)* node, atrial internodal pathways, and ventricular bundle branches.



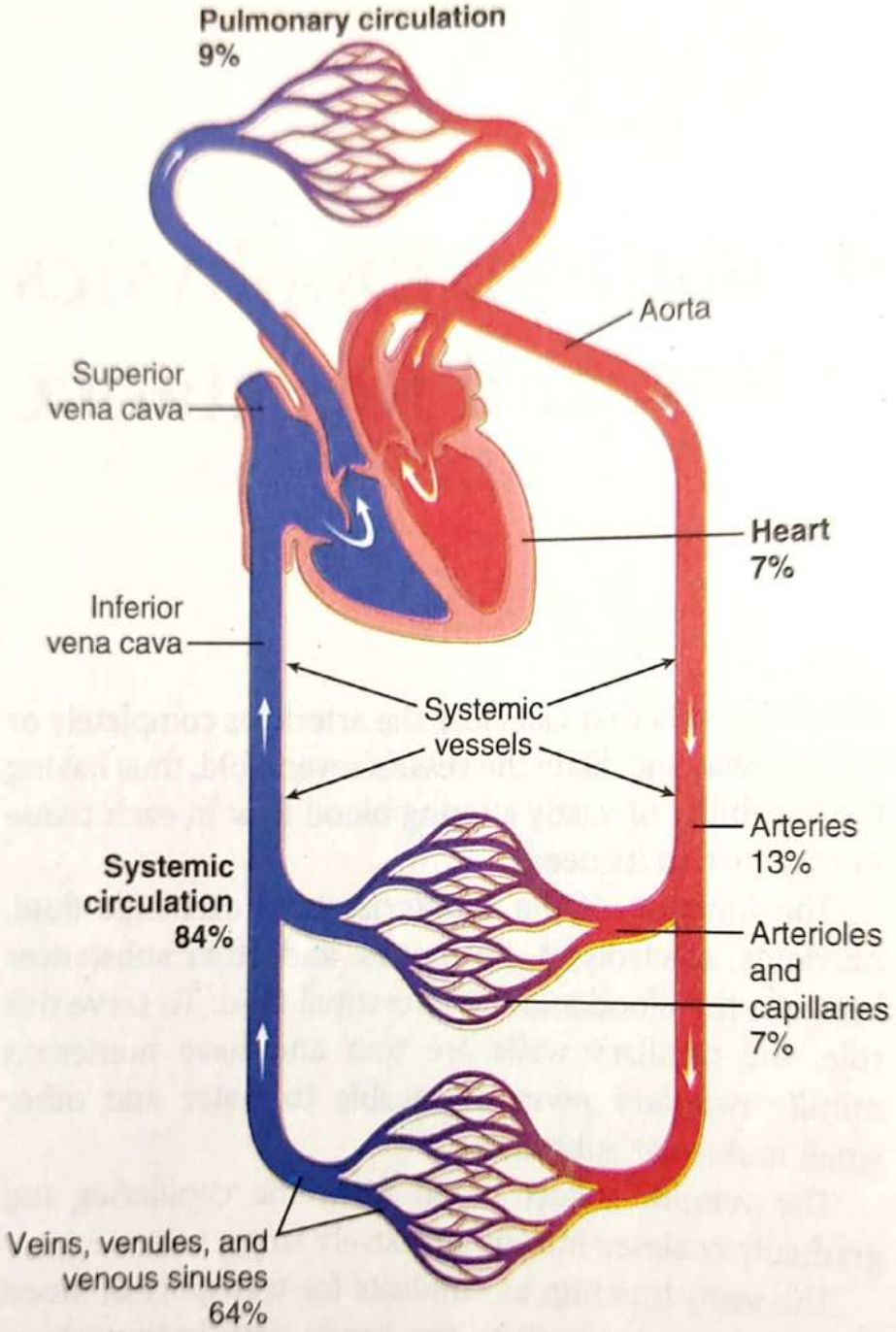


Figure 14-1. Distribution of blood (in percentage of total blood) in the different parts of the circulatory system.

Vessel	Cross-Sectional Area (cm^2)
Aorta	2.5
Small arteries	20
Arterioles	40
Capillaries	2500
Venules	250
Small veins	80
Venae cavae	8

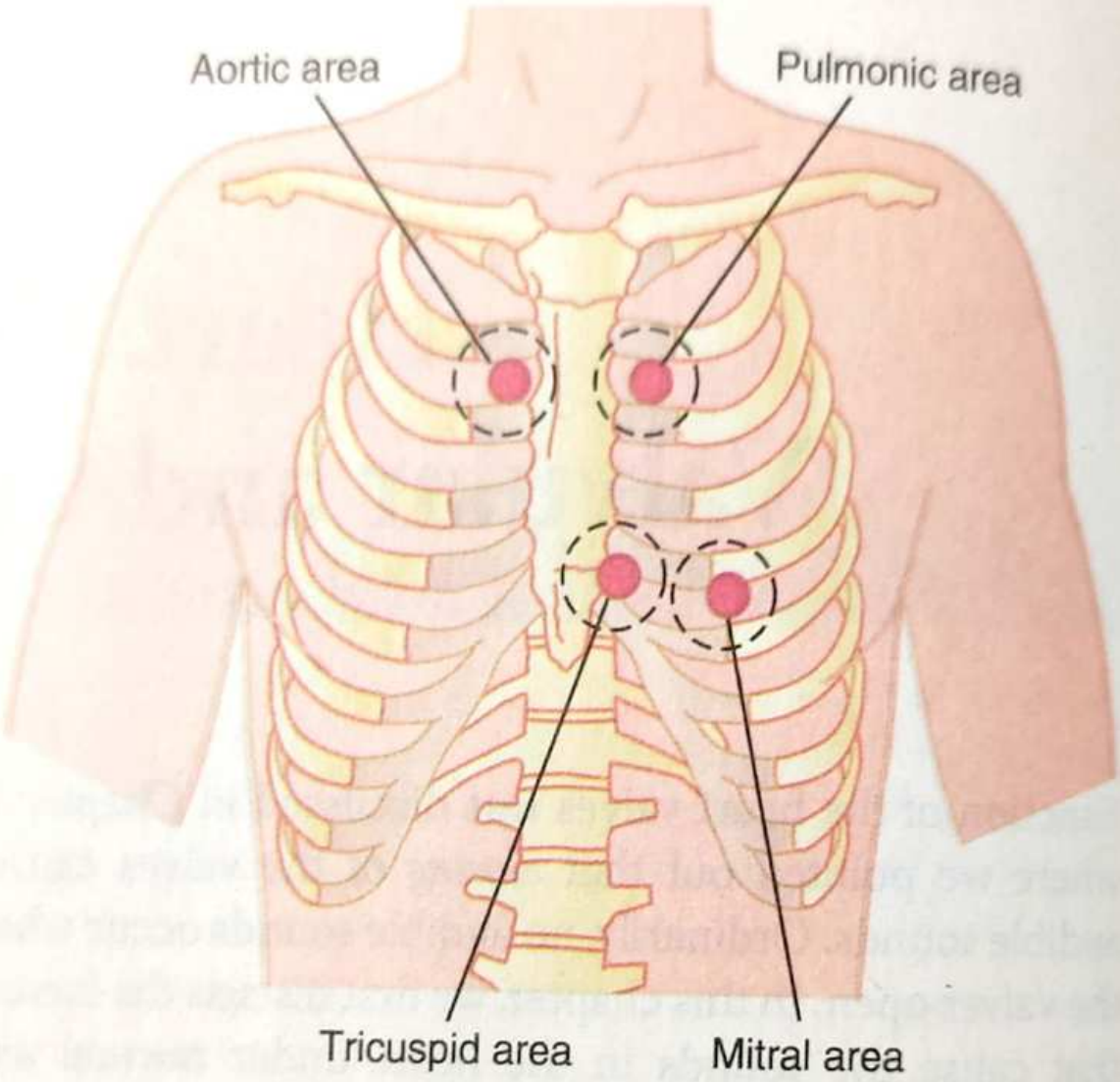


Figure 23-2. Chest areas from which sound from each valve is best heard.

and increased resistance to ventricular filling. For example, a fourth heart sound is often heard in older patients with left ventricular hypertrophy.

Chest Surface Areas for Auscultation of Normal Heart Sounds. Listening to the sounds of the body, usually with the aid of a stethoscope, is called *auscultation.* **Figure 23-2** shows the areas of the chest wall from which the different heart valvular sounds can best be heard. Although the sounds from all the valves can be heard from all these areas, the cardiologist distinguishes sounds from the

Date:

- Homeostasis: The maintenance of a constant environment in the body. Homeo ⇒ unchanging
stasis ⇒ standing

OR.

The maintenance of the internal environment with narrow range.

Body cells = 100 Trillions
- 25 Trillions RBCs
- 75 Trillions other cells.

BODY
- 40% solids.
- 60% Fluids.
  - Intracellular (2/3) (K^+, Mg^{2+}, PO_4^{3-})
  - Extracellular (1/3) (Na^+, Cl^-, HCO_3^-)
    - Interstitial (3/4 ECF) Fluid.
    - Blood (1/4 of ECF)

organ involves in Homeostasis: Kidneys = Na, Cl^-, HCO_3^-
lungs = CO_2, O_2

SYSTEMS ASSOCIATED WITH HOMEO: stomach, Liver etc = sugar, fatty acid, Amino acid

* Nervous system Associated with Homeostasis:
  - sensory system → sensory Receptors.
  - Center → Brain & spinal cord.
  - Motor system. → Effectors.

* Autonomic N.S
(Breathing, Heart rate, Blood Pressure).

Product of BAREENA

Date:

* Hormonal System:	* Integumentary System
8 - Glands.	
• Thyroid = ↑ rate of Rx.	- Barriers.
Insulin = Glucose metabolism.	- Temp regulation
• Adreno = Protein mulabolism	• waste excretion
corticotropic	• cushion.
Parathyroid = Calcium Balance	
Hormone	

• Immune System:

↳ WBCs + Thymus + Lymphnodes → spleen.
↳ kills pathogens, virus, Bacteria etc

part II

CONTROL SYSTEM OF BODY:

FEEDBACK SYSTEM:

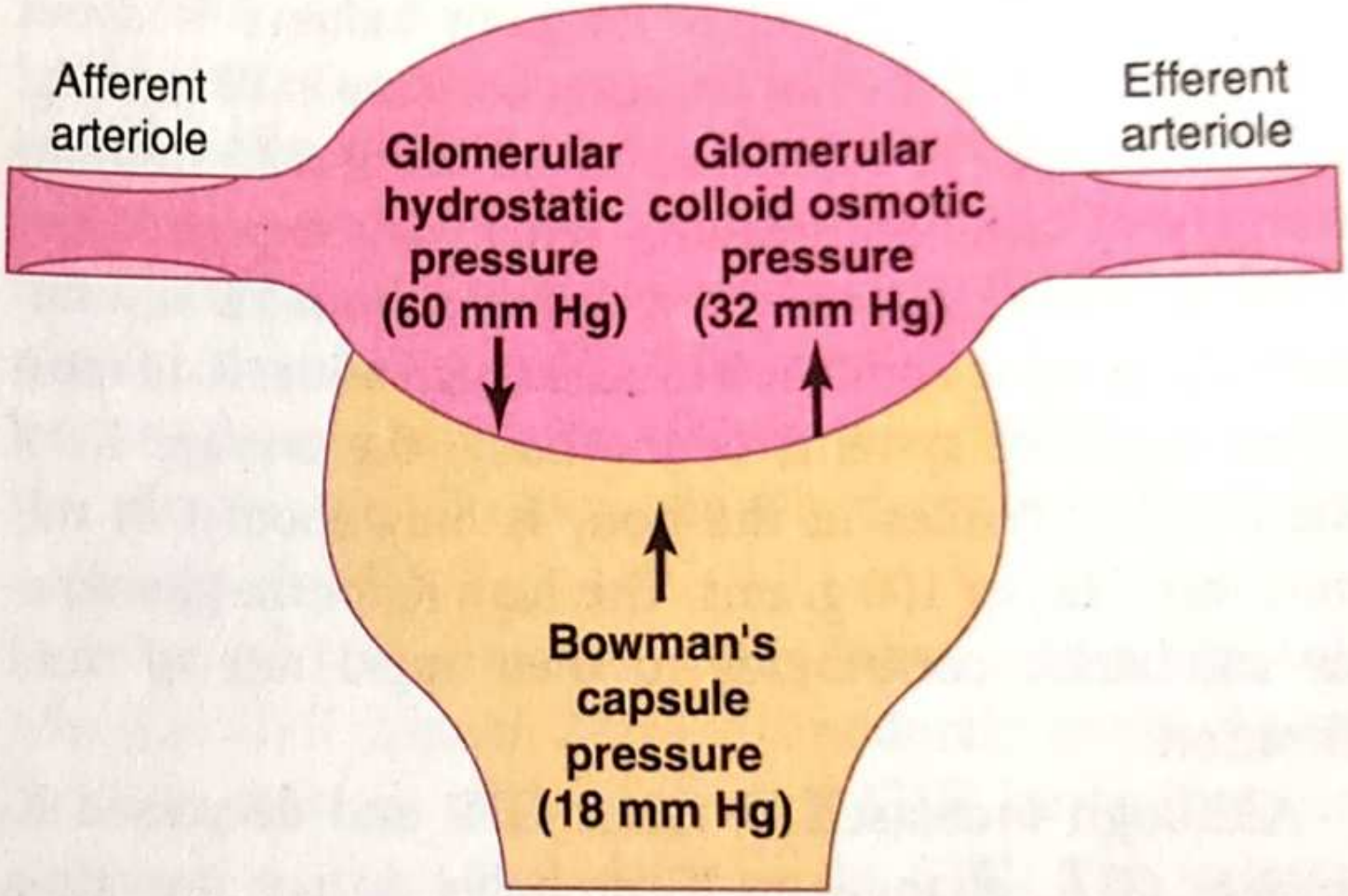
+ve Feed Back	-ve feed Back.
Blood vessel ruptured	↑ Body temp → sweating → ↓ Body temp
clotting ↓ enzyme activate	↑ CO_2 → Hyperventilation → CO_2 ↓
more activation of	
clotting enzyme.	
And.	GAIN:
child birth → Uterine	Degree of effectiveness to maintain
Contraction → Baby's	constant conditions.
Head stimulates cervix	Gain = Correction = Improvement Done
→ more Uterine Contr.	Error Improvement not Done
	• Greater the gain, Greater the control
	system is accurate

Product of BAREENA

Date:

FEEDFORWARD SYSTEM:-

Rapidly occuring movement of body → Brain sends Feedforward signal to muscle for upcoming movement by learning from the previous incorrectly performed movement.



Net filtration pressure (10 mm Hg)	=	Glomerular hydrostatic pressure (60 mm Hg)	−	Bowman's capsule pressure (18 mm Hg)	−	Glomerular oncotic pressure (32 mm Hg)

Figure 27-4. Summary of forces causing filtration by the glomerular capillaries. The values shown are estimates for healthy humans.

(Figure
oles increases both glomerular hydrostatic pressure and GFR.

Constriction of the efferent arterioles increases the resistance to outflow from the glomerular capillaries. This mechanism raises glomerular hydrostatic pressure, and as long as the increase in efferent resistance does not reduce renal blood flow too much, GFR increases slightly (see Figure 27-6). However, because efferent arteriolar constriction also reduces renal blood flow, filtration fraction

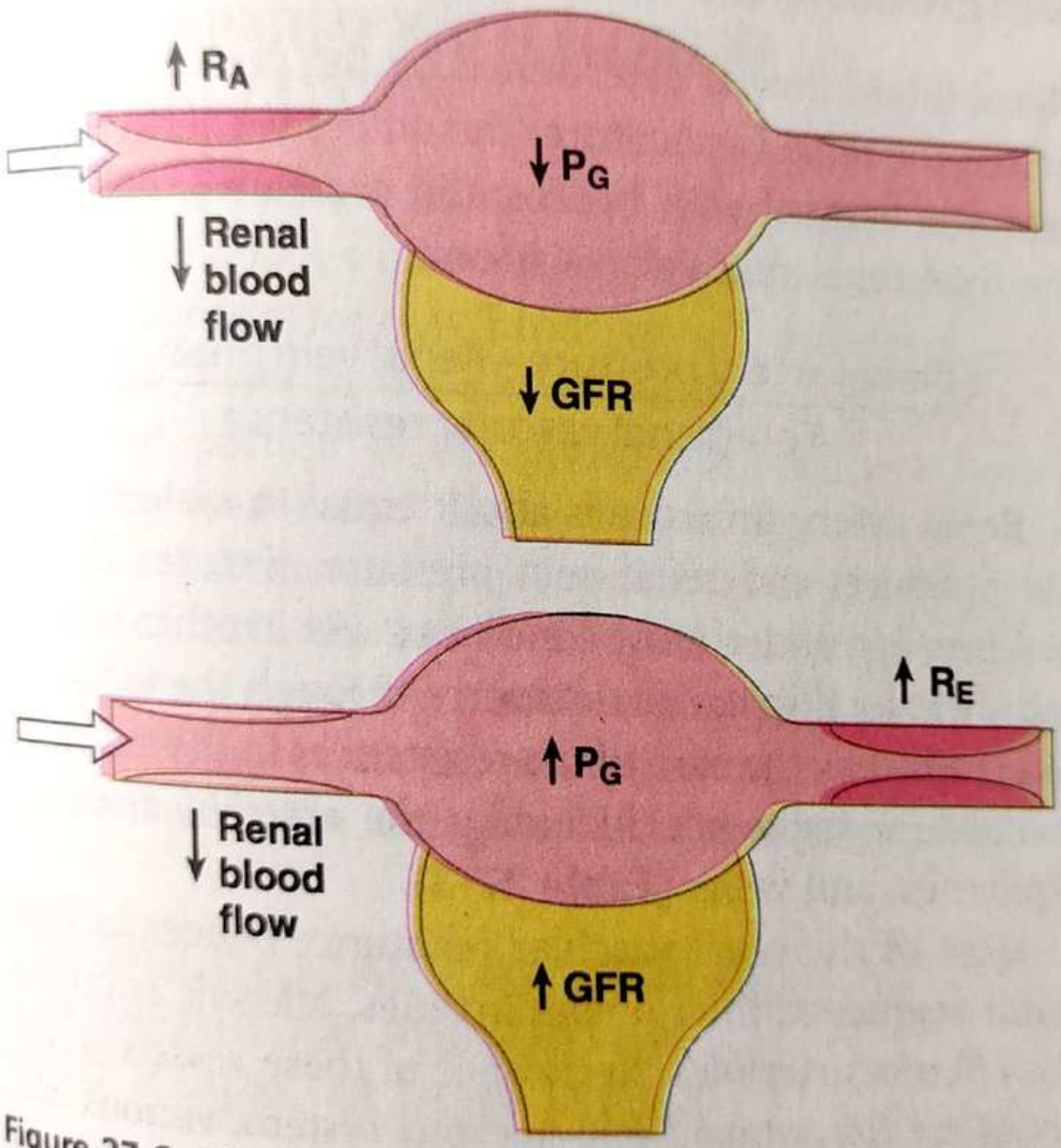


Figure 27-6. Effect of increases in afferent arteriolar resistance (R_A, top panel) or efferent arteriolar resistance (R_E, bottom panel) on renal blood flow, glomerular hydrostatic pressure (P_G), and glomerular filtration rate (GFR).

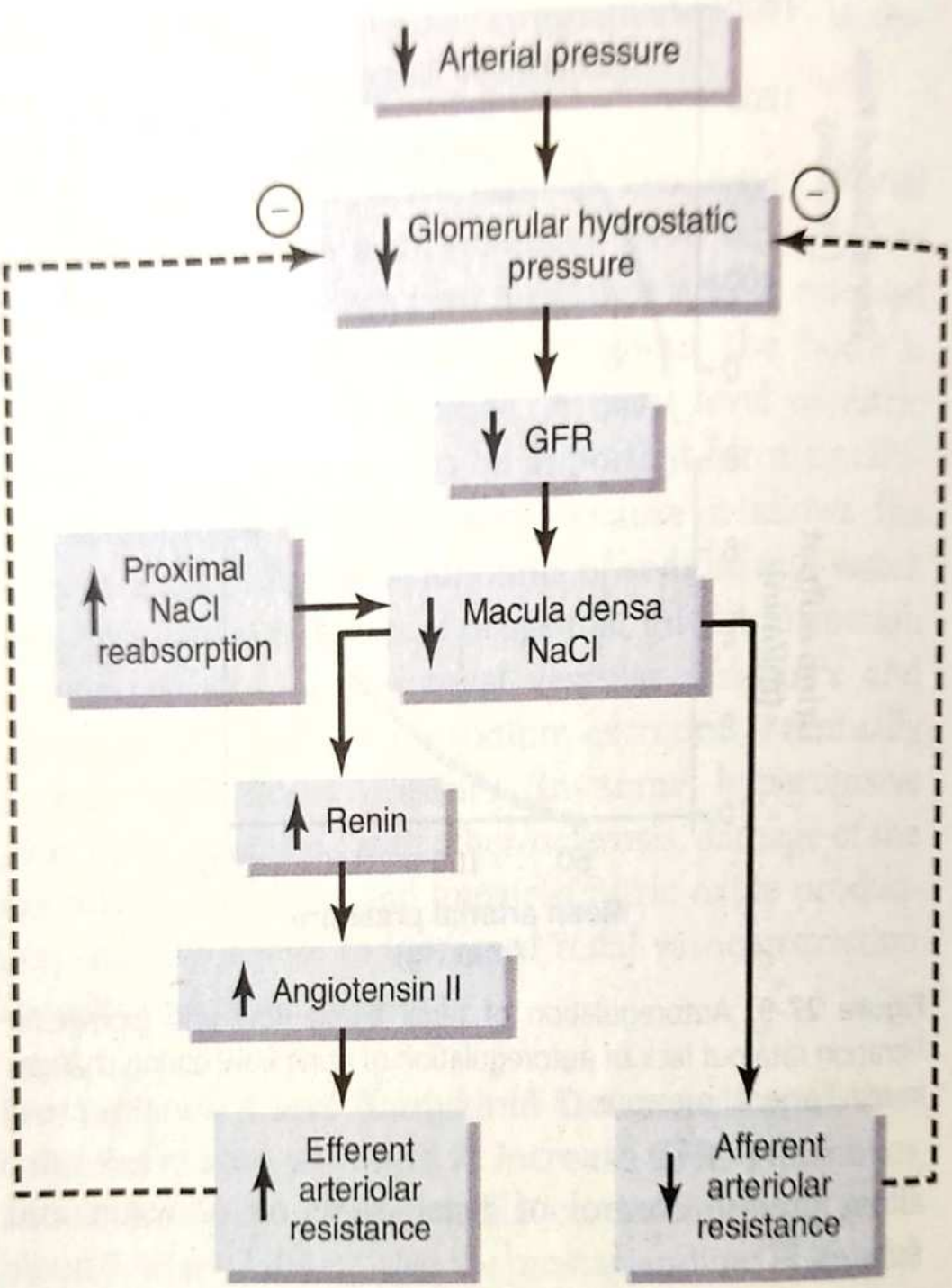


Figure 27-11. Macula densa feedback mechanism for autoregulation of glomerular hydrostatic pressure and glomerular filtration rate *(GFR)* during decreased renal arterial pressure.

preferential constrictor action of angiotensin II on efferent arterioles helps prevent serious reductions in glomerular hydrostatic pressure and GFR when renal perfusion pressure falls below normal. Administration of drugs that block the formation of angiotensin II (angiotensin-converting enzyme inhibitors) or that block the action of angiotensin II (angiotensin II receptor antagonists) may cause greater reductions in GFR than usual when the renal arterial re falls below normal. Therefore, an important cation of using these drugs to treat patients who

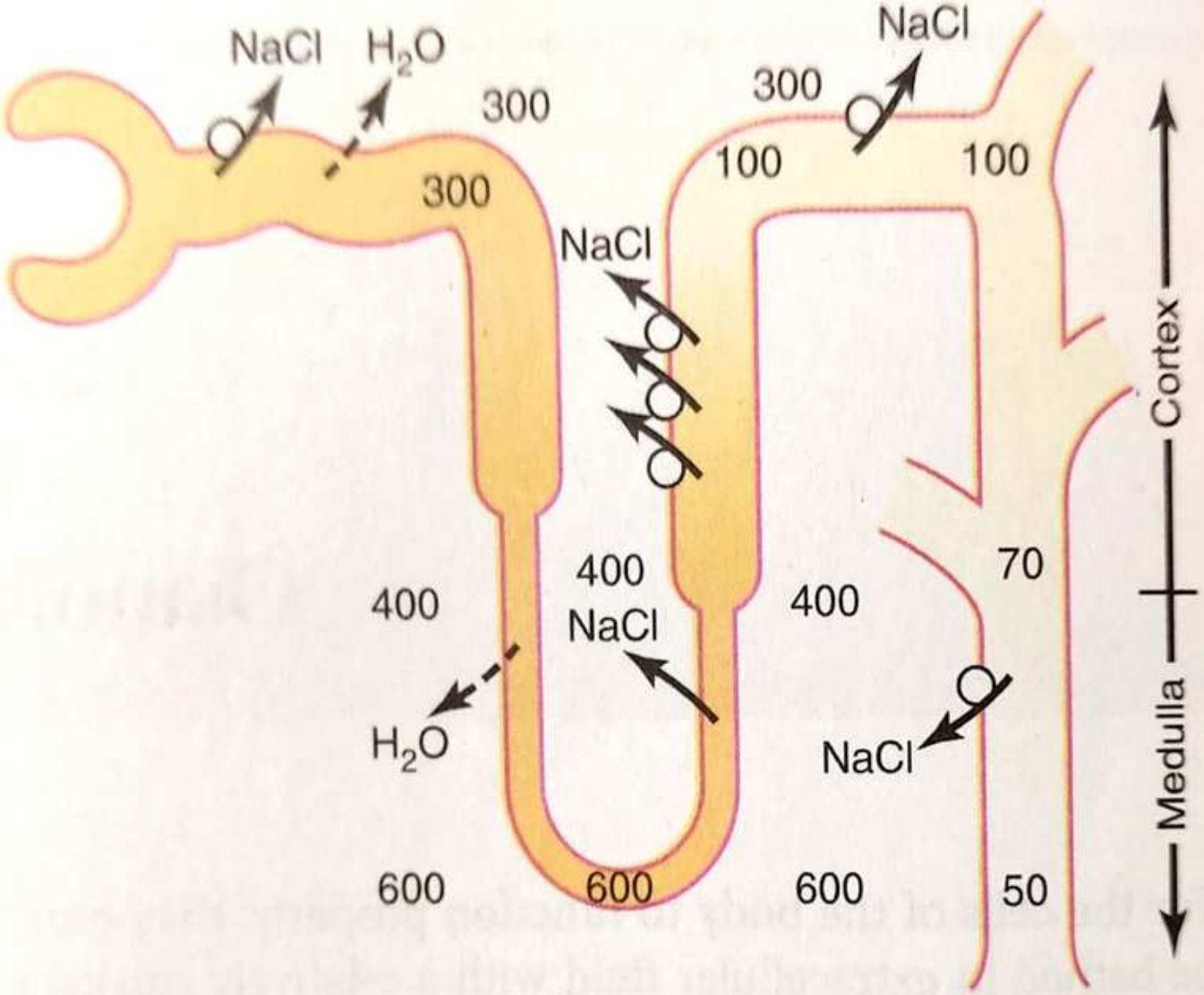


Figure 29-2. Formation of dilute urine when antidiuretic hormone (ADH) levels are very low. Note that in the ascending loop of Henle, the tubular fluid becomes very dilute. In the distal tubules and collecting tubules, the tubular fluid is further diluted by the reabsorption of sodium chloride and the failure to reabsorb water when ADH levels are very low. The failure to reabsorb water and continued reabsorption of solutes lead to a large volume of dilute urine. (Numerical values are in milliosmoles per liter.)

especially in the thick segment, sodium, potassium, and chloride are avidly reabsorbed. However, this portion of the tubular segment is impermeable to water, even in the presence of large amounts of ADH. Therefore, the tubular fluid becomes more dilute as it flows up the ascending loop of Henle into the early distal tubule, with the osmolarity decreasing progressively to about 100 mOsm/L by

When the tubular fluid leaves the loop of Henle and flows into the distal convoluted tubule in the renal cortex, the fluid is dilute, with an osmolarity of only about 100 mOsm/L (**Figure 29-5**). The early distal tubule further dilutes the tubular fluid because this segment, like the ascending loop of Henle, actively transports sodium chloride out of the tubule but is relatively impermeable to water.

As fluid flows into the cortical collecting tubule, the amount of water reabsorbed is critically dependent on the plasma concentration of ADH. In the absence of ADH, this segment is almost impermeable to water and fails to reabsorb water but continues to reabsorb solutes and

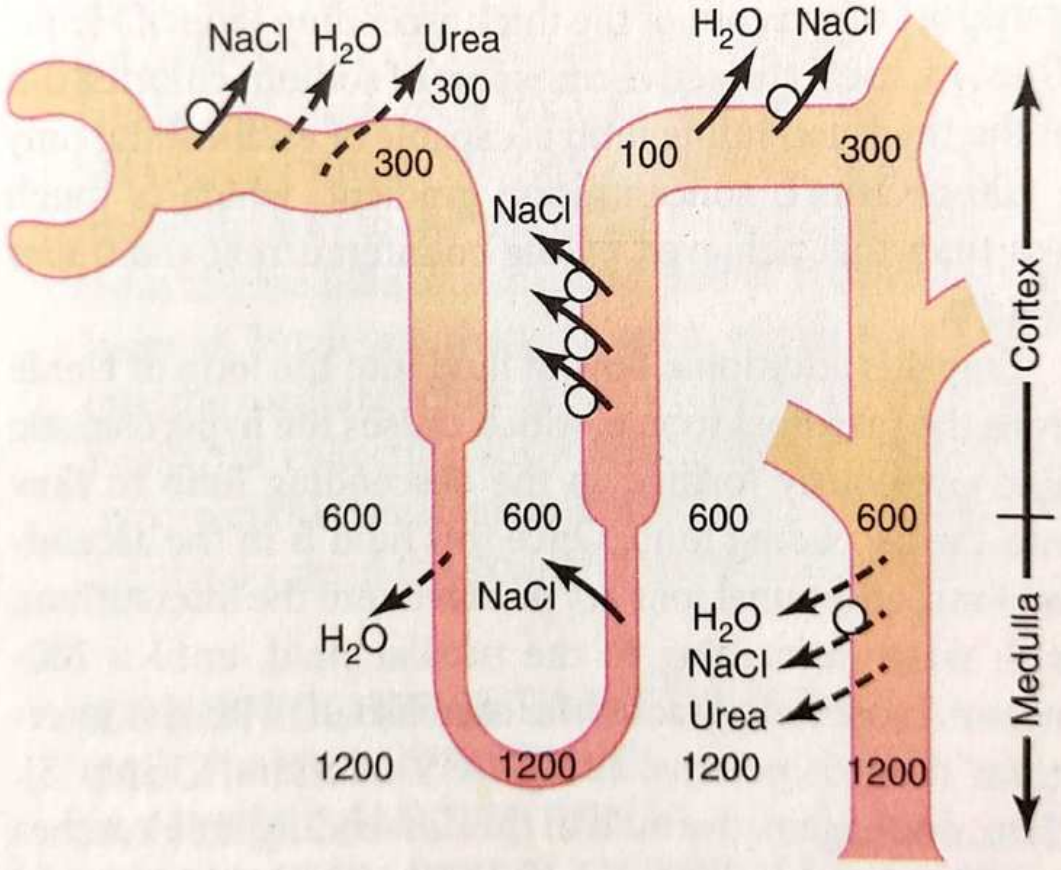


Figure 29-5. Formation of a concentrated urine when antidiuretic hormone (ADH) levels are high. Note that the fluid leaving the loop of Henle is dilute but becomes concentrated as water is absorbed from the distal tubules and collecting tubules. With high ADH levels, the osmolarity of the urine is about the same as the osmolarity of the renal medullary interstitial fluid in the papilla, which is about 1200 mOsm/L. (Numerical values are in milliosmoles per liter.)

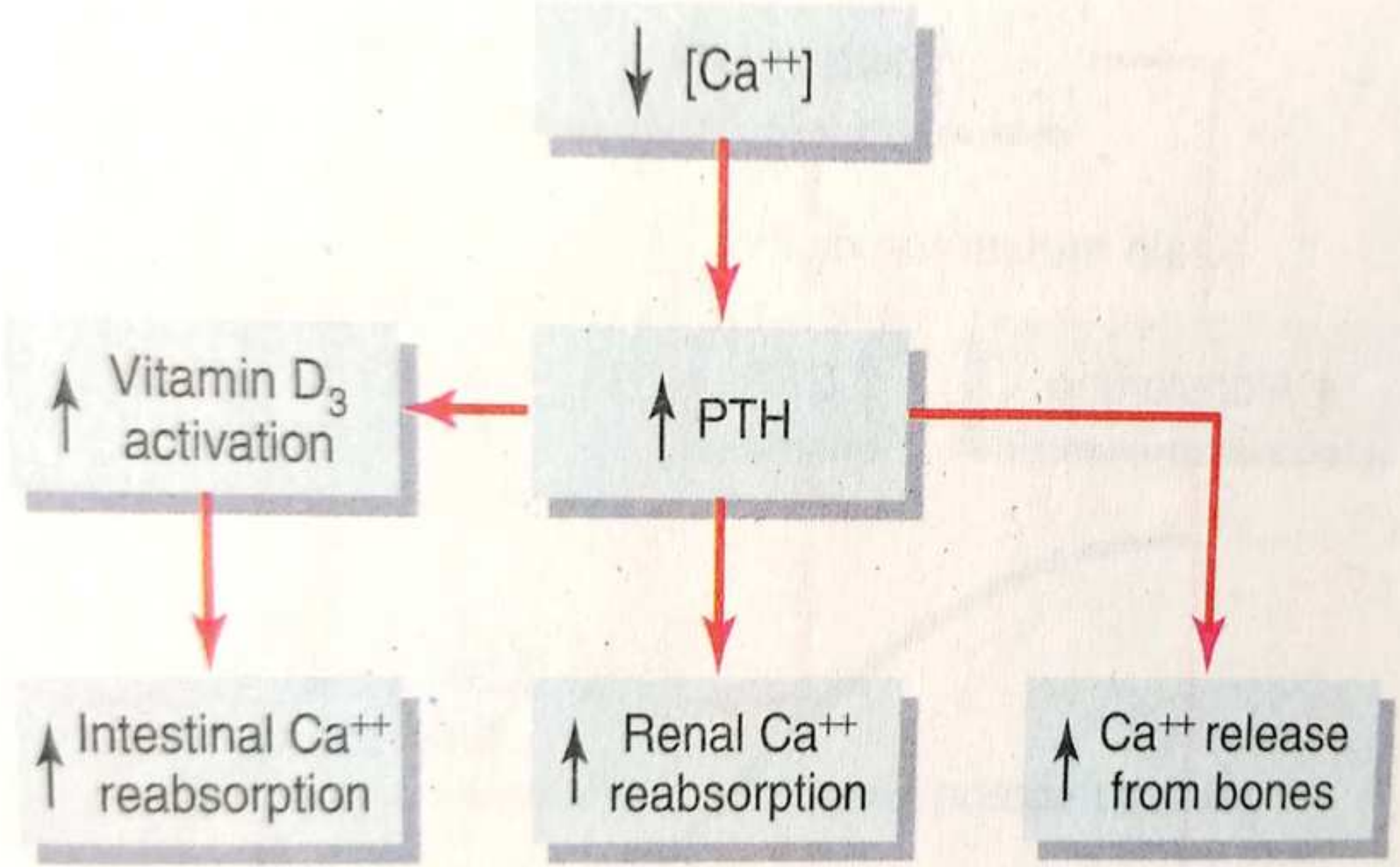


Figure 30-11. Compensatory responses to decreased plasma ionized calcium concentration mediated by parathyroid hormone *(PTH)* and vitamin D.

Figure 32-1. Sodium excretion and extracellular fluid volume during diuretic administration. The immediate increase in sodium excretion is accompanied by a decrease in extracellular fluid volume. If sodium intake is held constant, compensatory mechanisms will eventually return sodium excretion to equal sodium intake, thus re-establishing sodium balance.

Acetazolamide inhibits the enzyme *carbonic anhydrase*, which is critical for the reabsorption of bicarbonate (HCO_3^-) in the proximal tubule, as discussed in Chapter 31. Carbonic anhydrase is abundant in the proximal

Table 32-1 Classes of Diuretics, Their Mechanisms of Action, and Tubular Sites of Action

Class of Diuretic	Mechanism of Action	Tubular Site of Action
Osmotic diuretics (mannitol)	Inhibit water and solute reabsorption by increasing osmolarity of tubular fluid	Mainly proximal tubules
Loop diuretics (furosemide, bumetanide)	Inhibit Na^+-K^+-Cl^- co-transport in luminal membrane	Thick ascending loop of Henle
Thiazide diuretics (hydrochlorothiazide, chlorthalidone)	Inhibit Na^+-Cl^- co-transport in luminal membrane	Early distal tubules
Carbonic anhydrase inhibitors (acetazolamide)	Inhibit H^+ secretion and HCO_3^- reabsorption, which reduces Na^+ reabsorption	Proximal tubules
Aldosterone antagonists (spironolactone, eplerenone)	Inhibit action of aldosterone on tubular receptor, decrease Na^+ reabsorption, and decrease K^+ secretion	Collecting tubules
Sodium channel blockers (triamterene, amiloride)	Block entry of Na^+ into Na^+ channels of luminal membrane, decrease Na^+ reabsorption, and decrease K^+ secretion	Collecting tubules

s are much more
transfusion reac-
antigens and the

OGENS

on the surfaces
man beings. It
s because they
at cause most
he way these
ave neither of
hey may have

somes, only one of these alleles is present on each of the two chromosomes in any individual. However, the presence of three different alleles means that there are six possible combinations of alleles, as shown in **Table 36-1**: OO, OA, OB, AA, BB, and AB. These combinations of

Table 36-1 Blood Types with Their Genotypes and Their Constituent Agglutinogens and Agglutinins

Genotypes	Blood Types	Agglutinogens	Agglutinins
OO	O	–	Anti-A and Anti-B
OA or AA	A	A	Anti-B
OB or BB	B	B	Anti-A
AB	AB	A and B	–

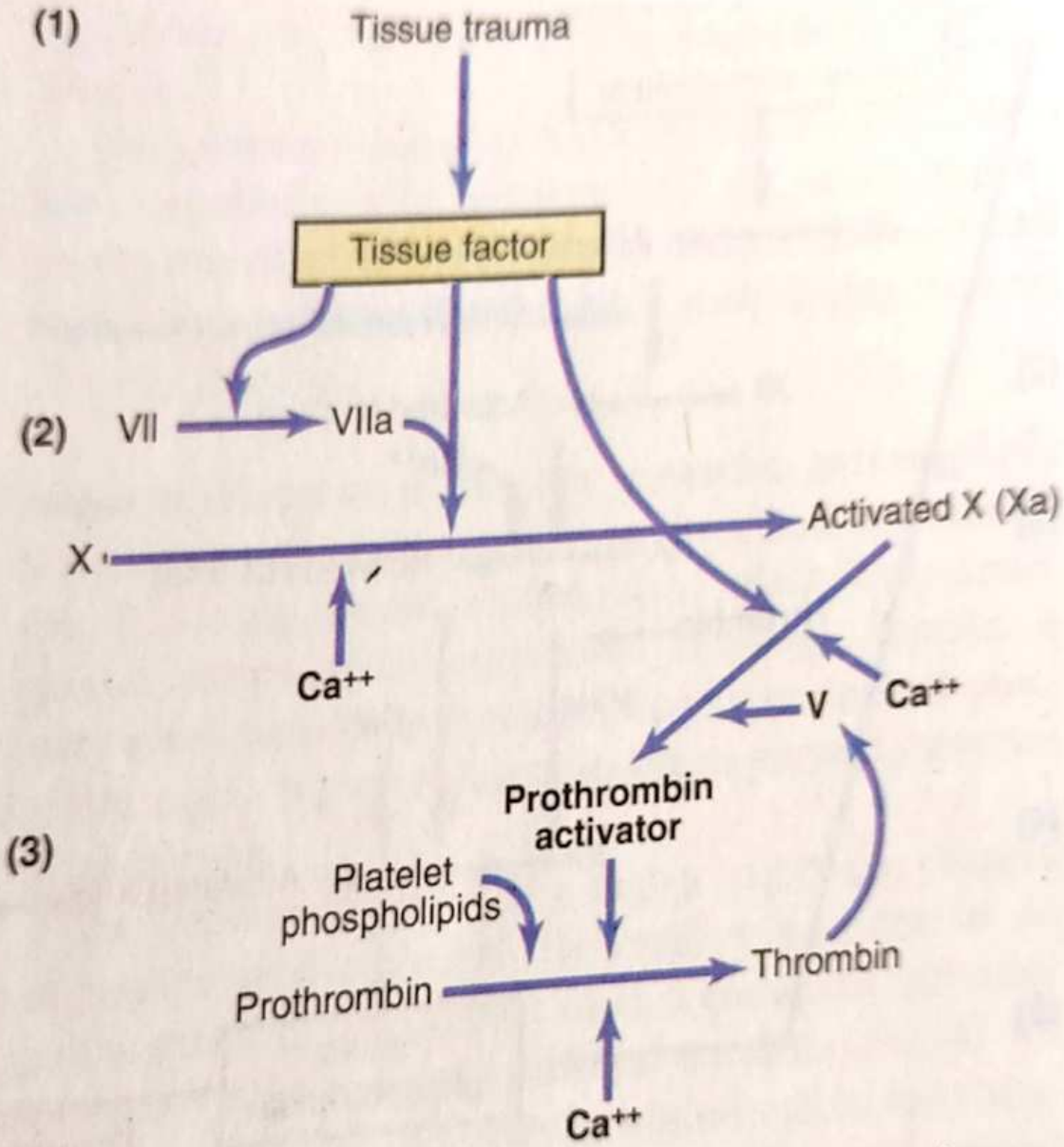
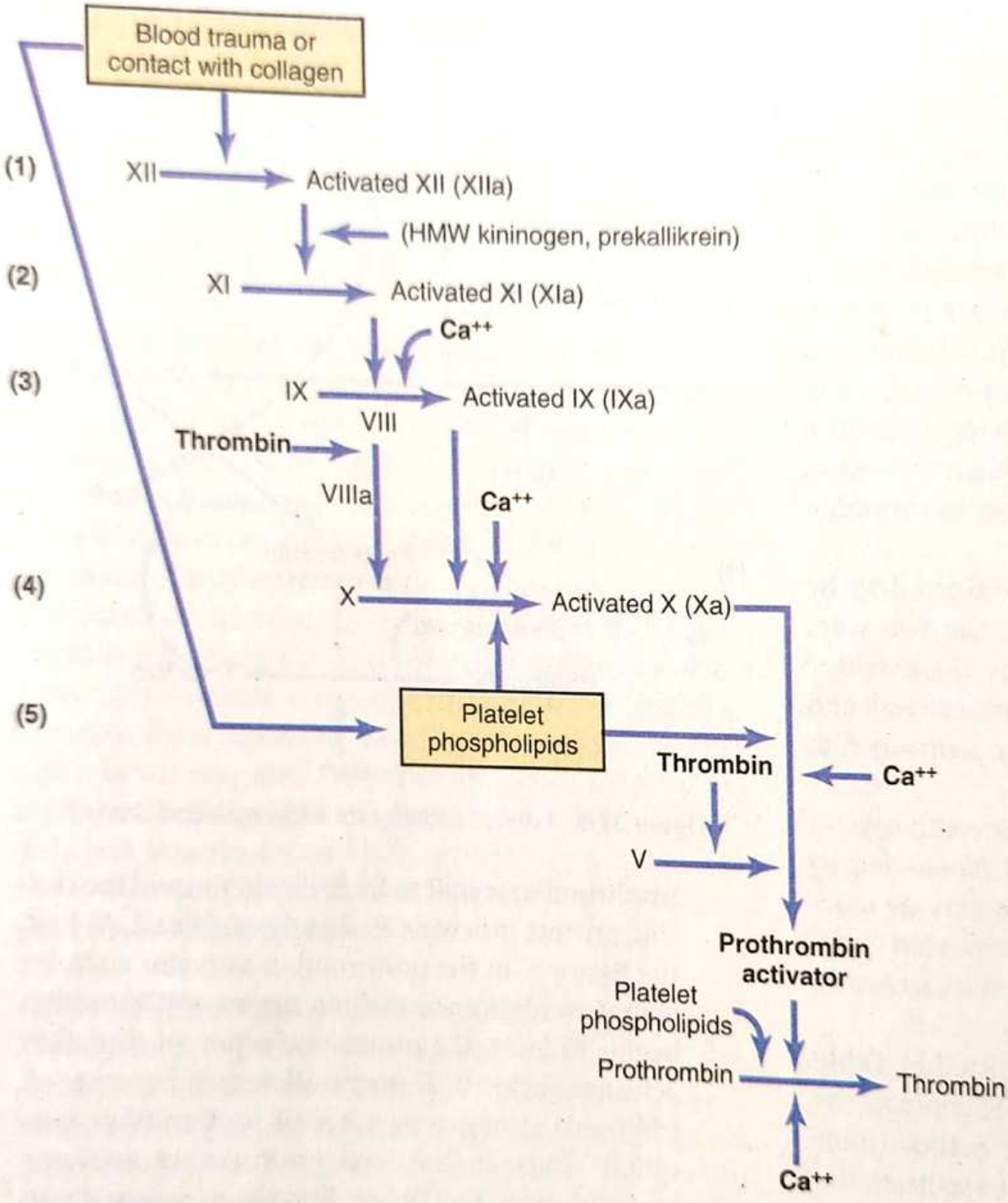


Figure 37-3. Extrinsic pathway for initiating blood clotting.

prothrombin is split to form thrombin, and the clotting process proceeds as already explained. At first, the Factor V in the prothrombin activator complex is inactive, but once clotting begins and thrombin begins to form, the proteolytic action of thrombin activates Factor V. This activation then becomes an additional strong accelerator of prothrombin activation. Thus, in the final prothrombin activator Factor X is the actual protease



either collagen or a wettable surface (or by damage in other ways), and this releases platelet phospholipids that contain the lipoprotein called *platelet factor 3*, which also plays a role in subsequent clotting reactions.

2. *Activation of Factor XI.* The activated Factor XII acts enzymatically on Factor XI to activate this factor as well, which is the second step in the intrinsic pathway. This reaction also requires *high-molecular-weight kininogen* and is accelerated by prekallikrein.

5. *Action of act*
activator—ro
pathway is th
pathway. Tha
Factor V and
form the com
prothrombin
seconds the
thrombin, the
ting process,

a is mainly in two
centration of $HPO_4^=$
entration of $H_2PO_4^-$
al quantity of phos-
so does the quantity
hate ions. Further-
lular fluid becomes
ease in $H_2PO_4^-$ and
posite occurs when
ine. These relations
cid-base balance in

hemically the exact
ne blood, ordinarily
pressed in terms of
(100 milliliters) of
rganic phosphorus
is about 4 mg/dl,

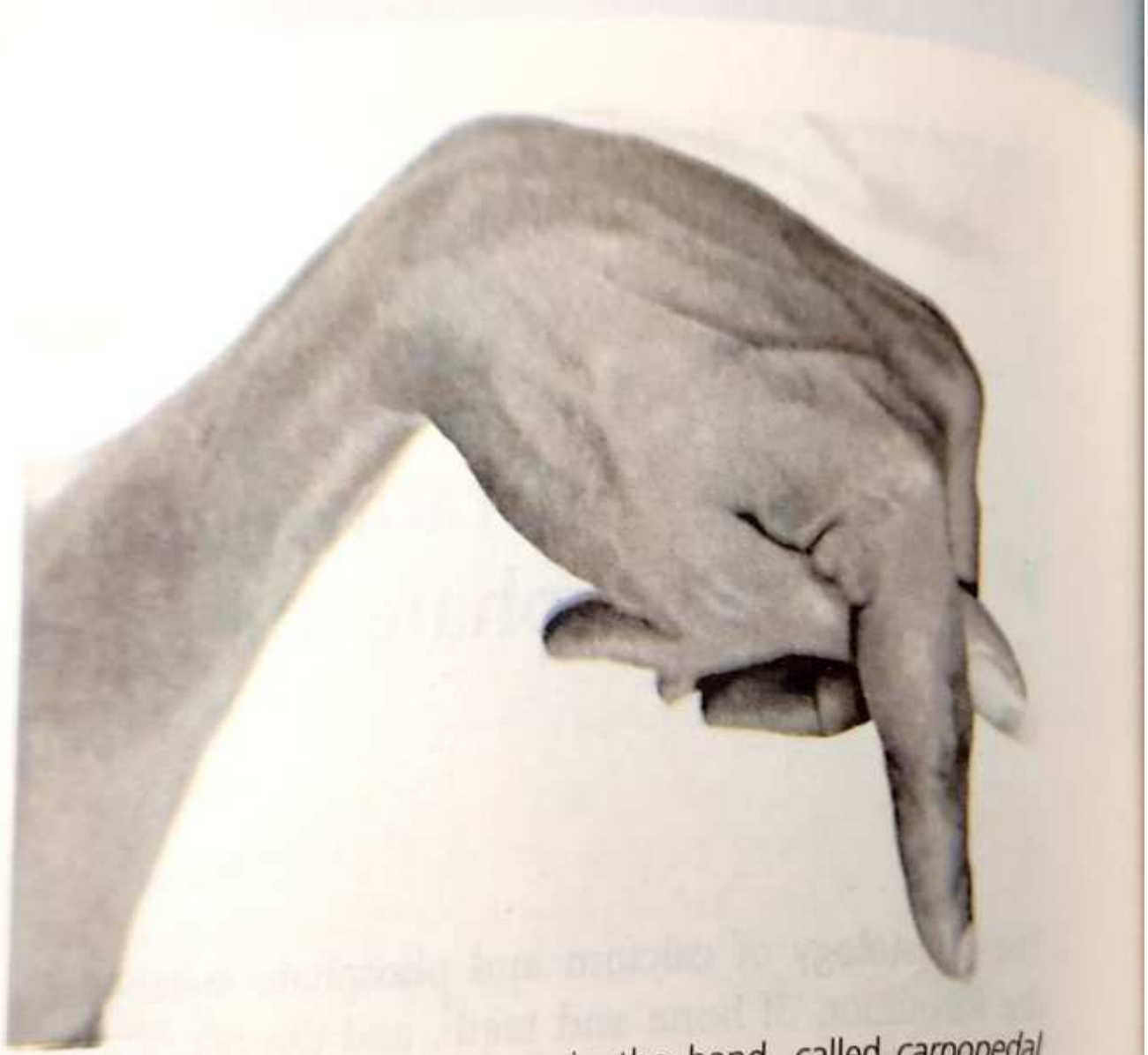


Figure 80-2. *Hypocalcemic tetany in the hand, called carpopedal spasm.*

calcium concentration, and it is usually lethal at about 4 mg/dl.

In laboratory animals, extreme hypocalcemia can

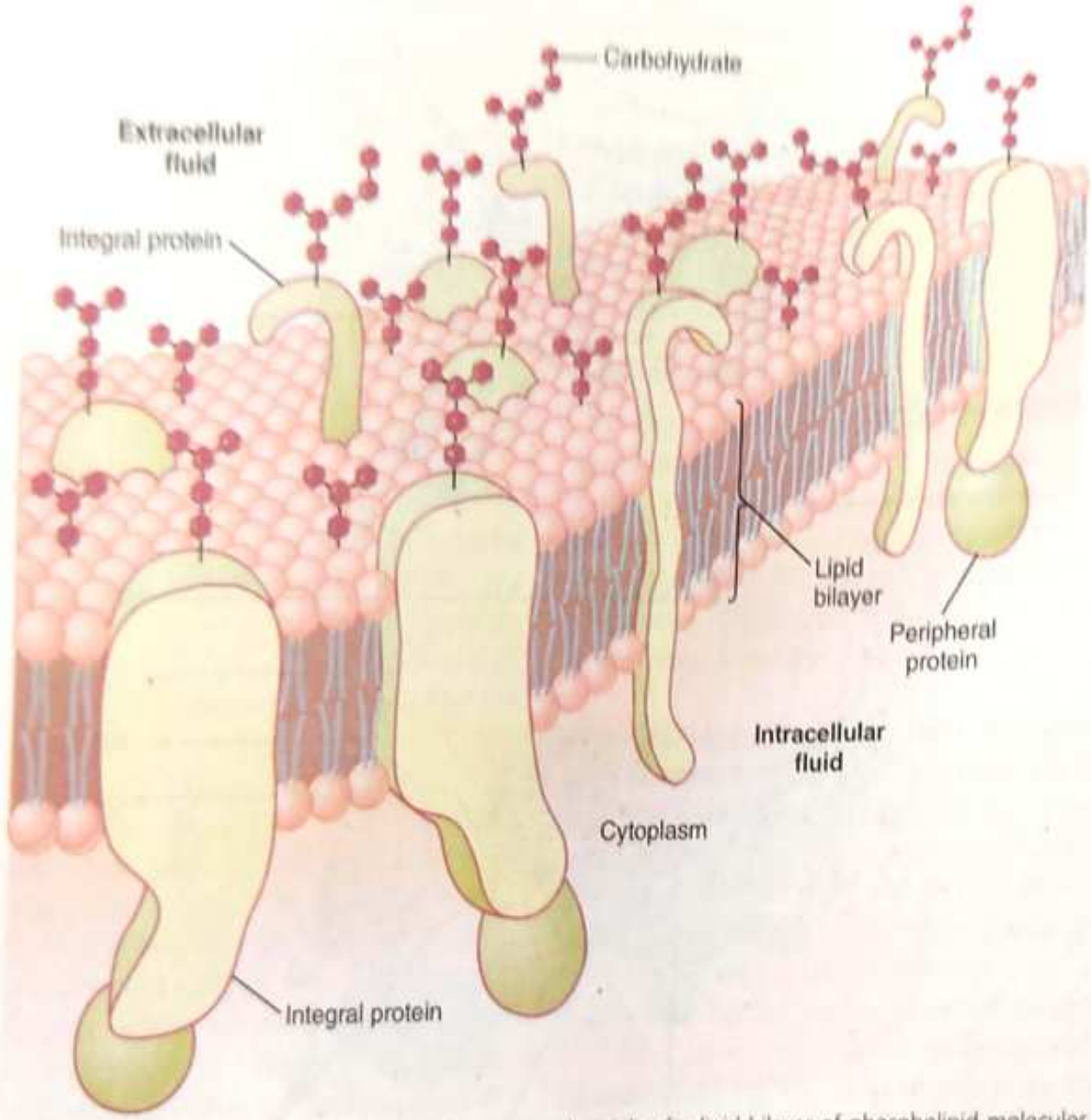
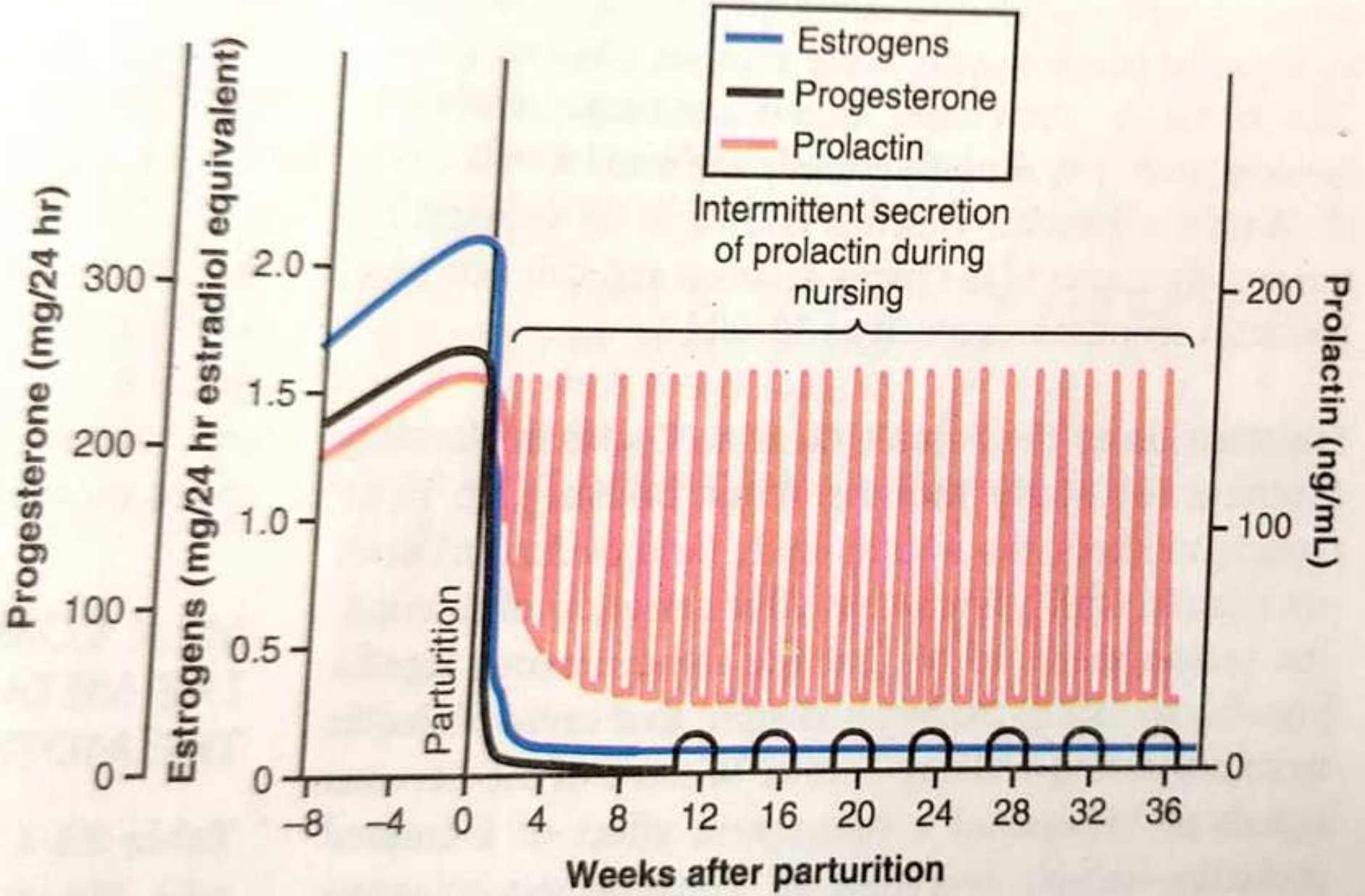


Figure 2-3. Structure of the cell membrane, showing that it is composed mainly of a lipid bilayer of phospholipid molecules, but with large numbers of protein molecules protruding through the layer. Also, carbohydrate moieties are attached to the protein molecules on the outside


Estrogens
Progesterone
Prolactin
Intermittent secretion of prolactin during nursing
Progesterone (mg/24 hr)
Estrogens (mg/24 hr estradiol equivalent)
Prolactin (ng/mL)
Parturition
Weeks after parturition
300
200
100
0
2.0
1.5
1.0
0.5
0
200
100
0
–8
–4
0
4
8
12
16
20
24
28
32
36

can breathe in, beginning at the normal expiratory level and distending the lungs to the maximum amount.

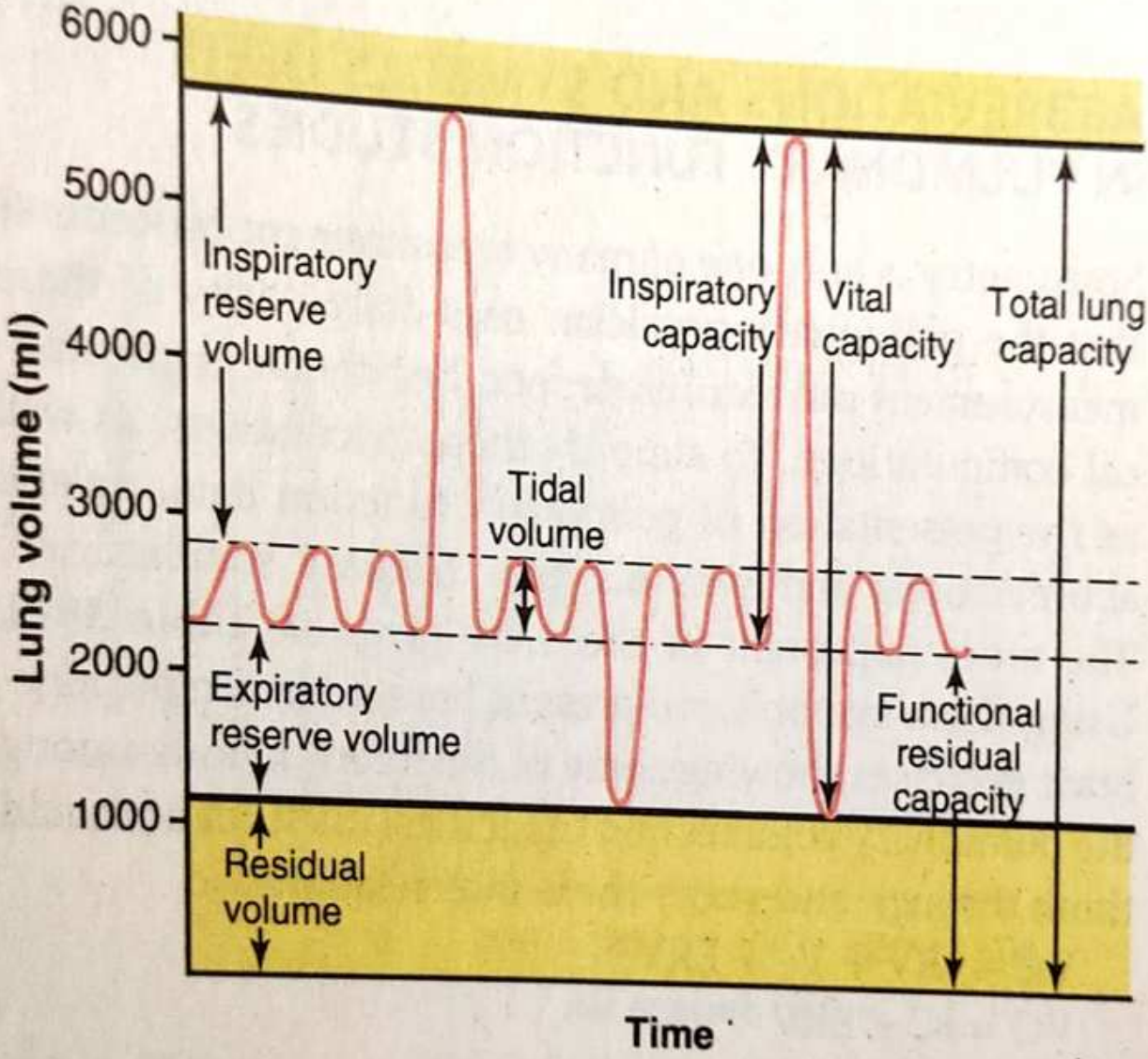


Figure 38-6. A diagram showing respiratory excursions during normal breathing and during maximal inspiration and maximal expiration.

Table 38-1 Average Pulmonary Volumes and Capacities for a Healthy, Young Adult Man

Pulmonary Volumes and Capacities	Normal Values (ml)
Volumes	
Tidal volume	500
Inspiratory reserve volume	3000
Expiratory volume	1100
Residual volume	1200
Capacities	
Inspiratory capacity	3500
Functional residual capacity	2300
Vital capacity	4600
Total lung capacity	5800

are called *inhibitory transmitters.*

gamma (γ) components that are attached to the alpha

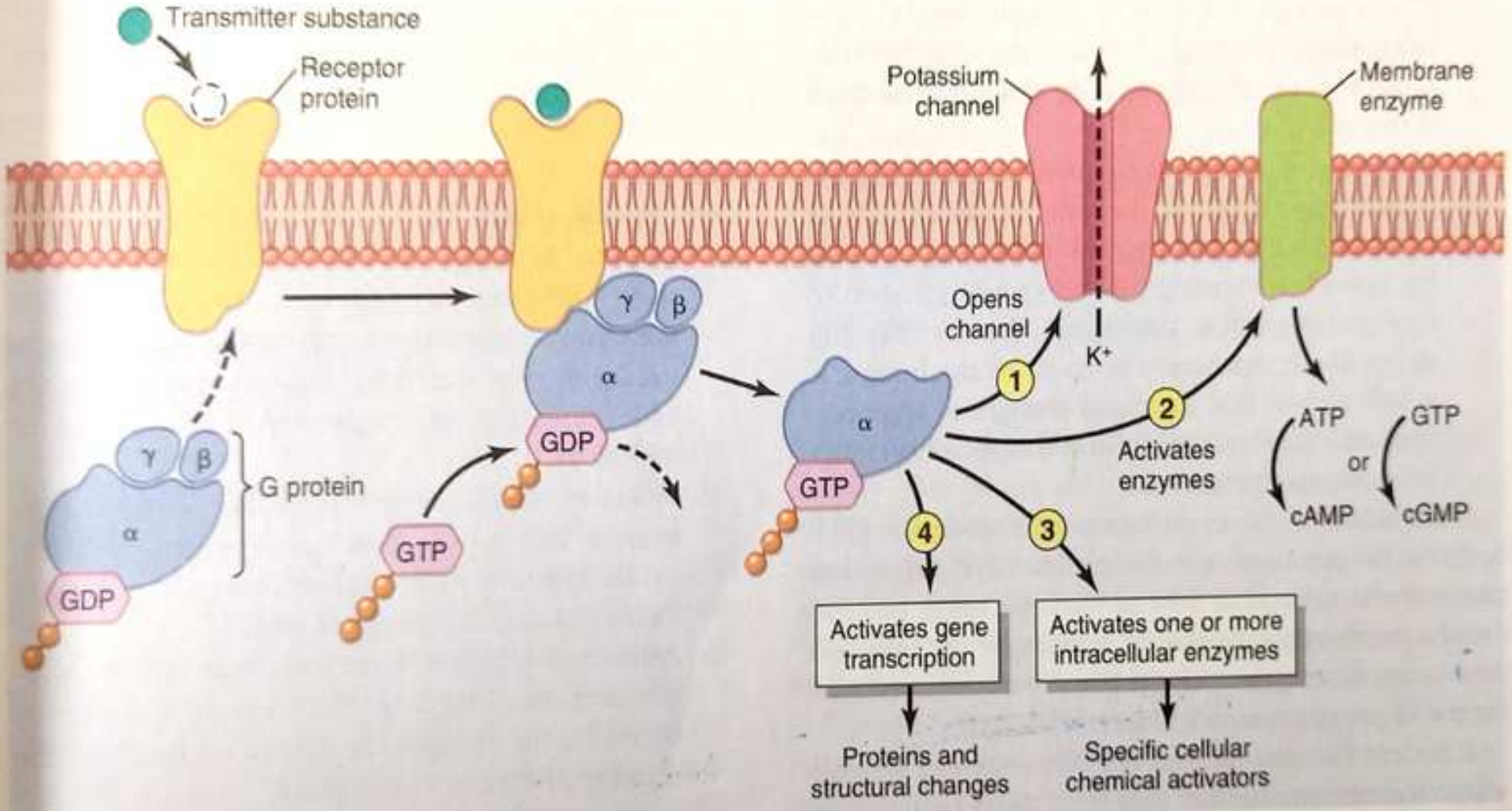


Figure 46-7. The "second messenger" system by which a transmitter substance from an initial neuron can activate a second neuron by first causing a transformational change in the receptor that releases the activated alpha (α) subunit of the G protein into the second neuron's cytoplasm. Four subsequent possible effects of the G protein are shown, including *1*, opening an ion channel in the membrane of the second neuron; *2*, activating an enzyme system in the neuron's membrane; *3*, activating an intracellular enzyme system; and/or *4*, causing gene transcription in the second neuron. Return of the G protein to the inactive state occurs when guanosine triphosphate *(GTP)* bound to the α subunit is hydrolyzed to guanosine diphosphate *(GDP)* and the β and γ subunits are reattached to the α subunit.

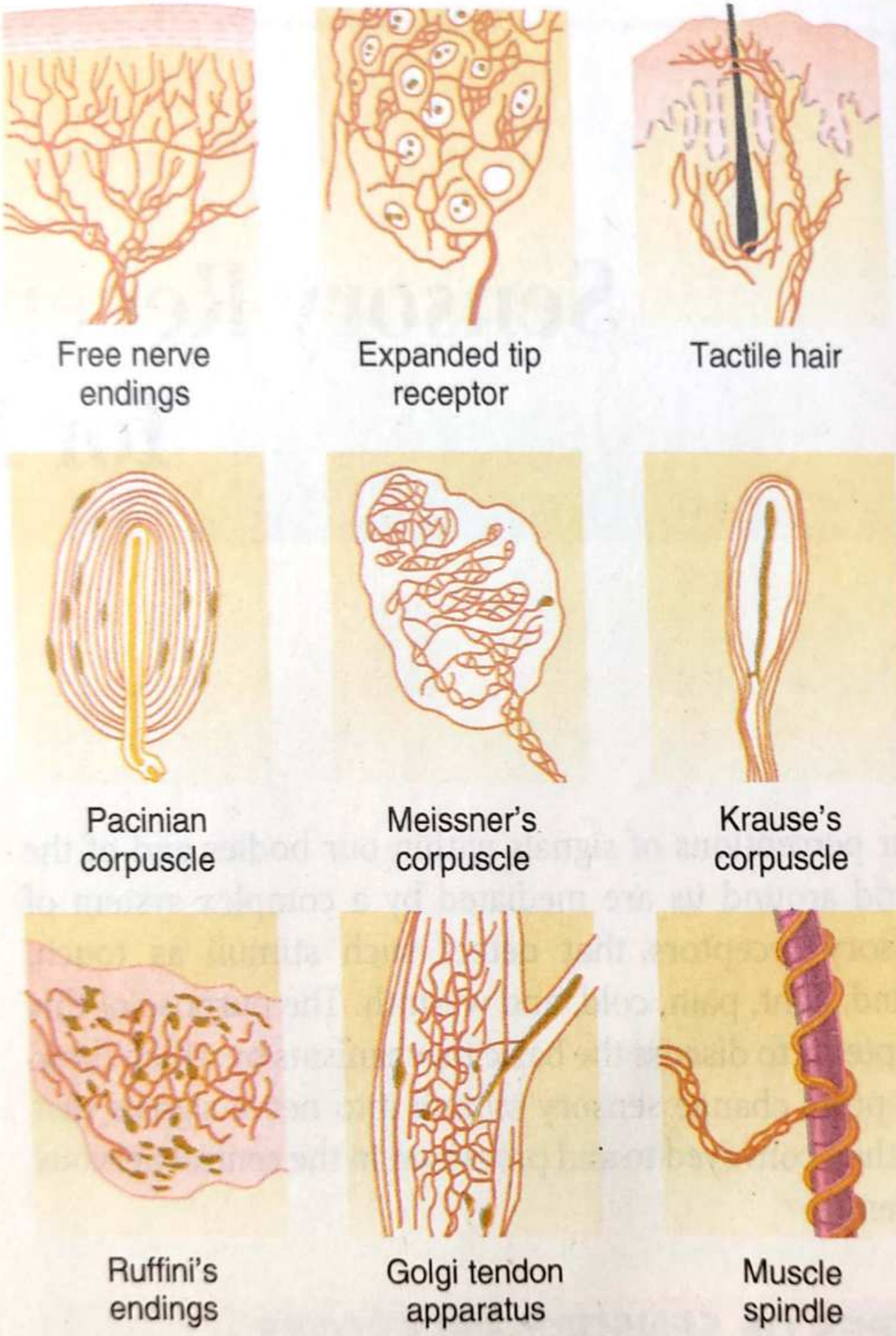


Figure 47-1. Several types of somatic sensory nerve endings.

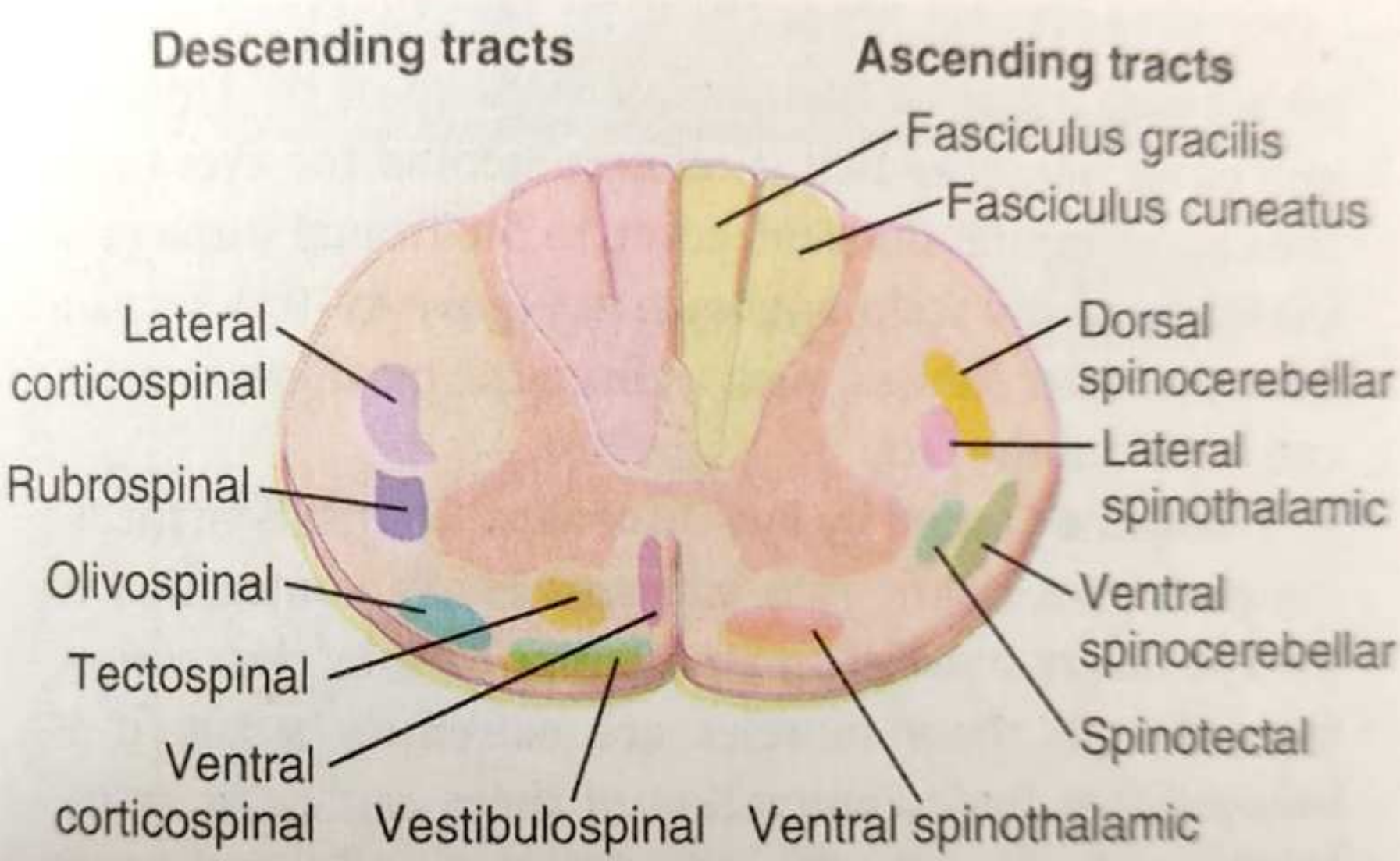


Figure 49-8. Cross section of the spinal cord, showing principal ascending tracts on the right and principal descending tracts on the left.

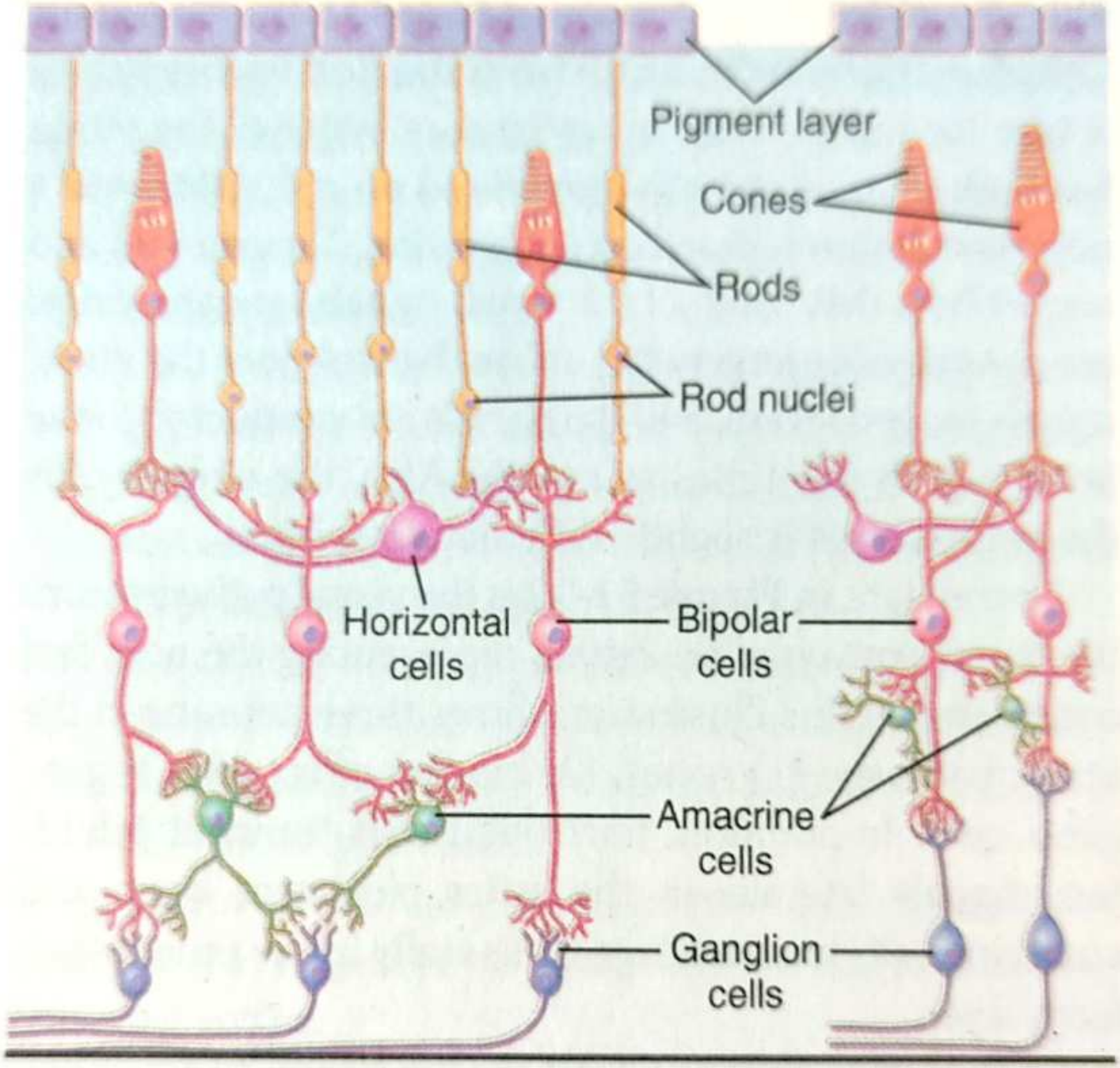


Figure 51-12. Neural organization of the retina, with the peripheral area to the left and the foveal area to the right.

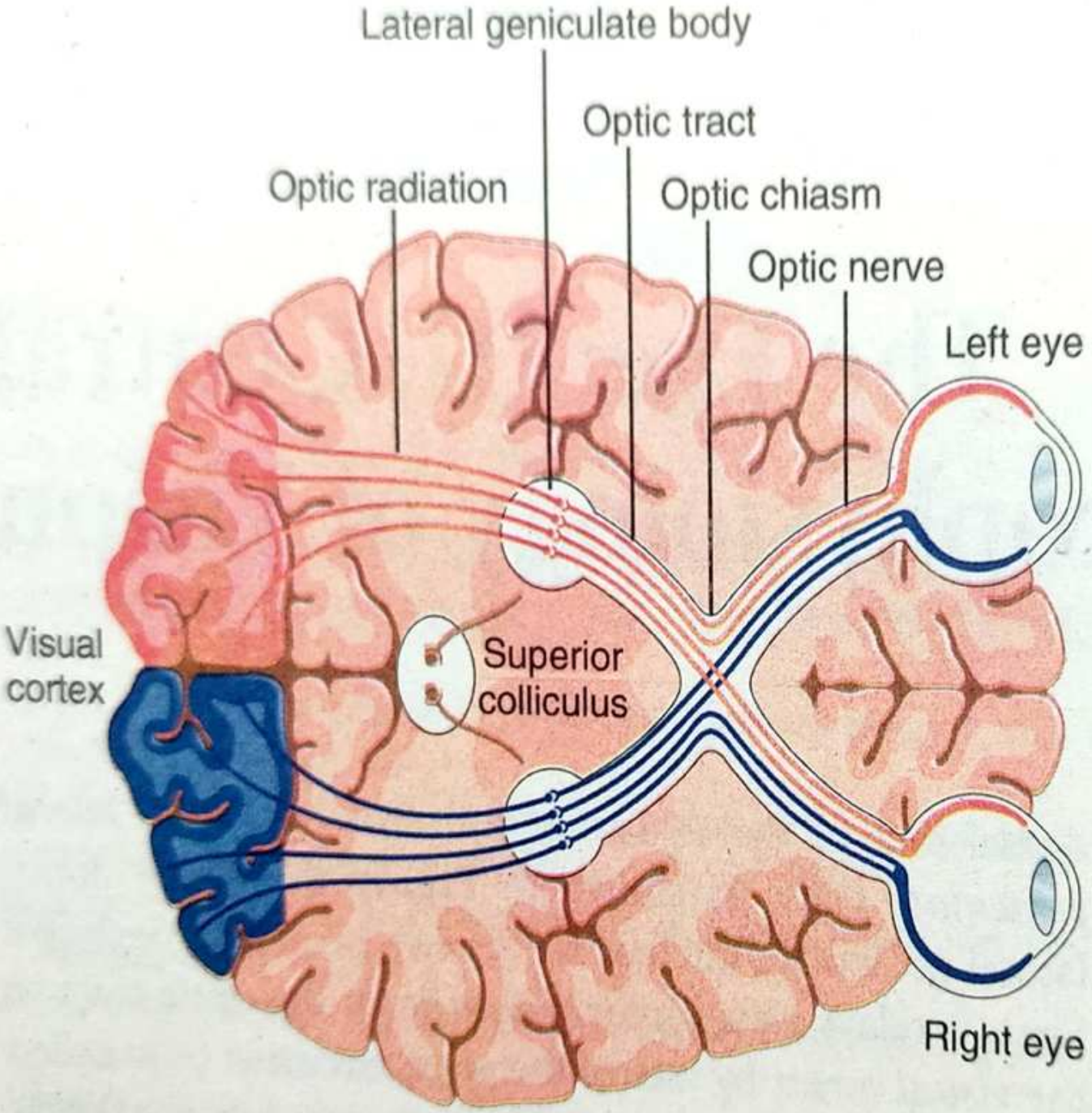


Figure 52-1. Principal visual pathways from the eyes to the visual cortex. *(Modified from Polyak SL: The Retina. Chicago: University of Chicago, 1941.)*

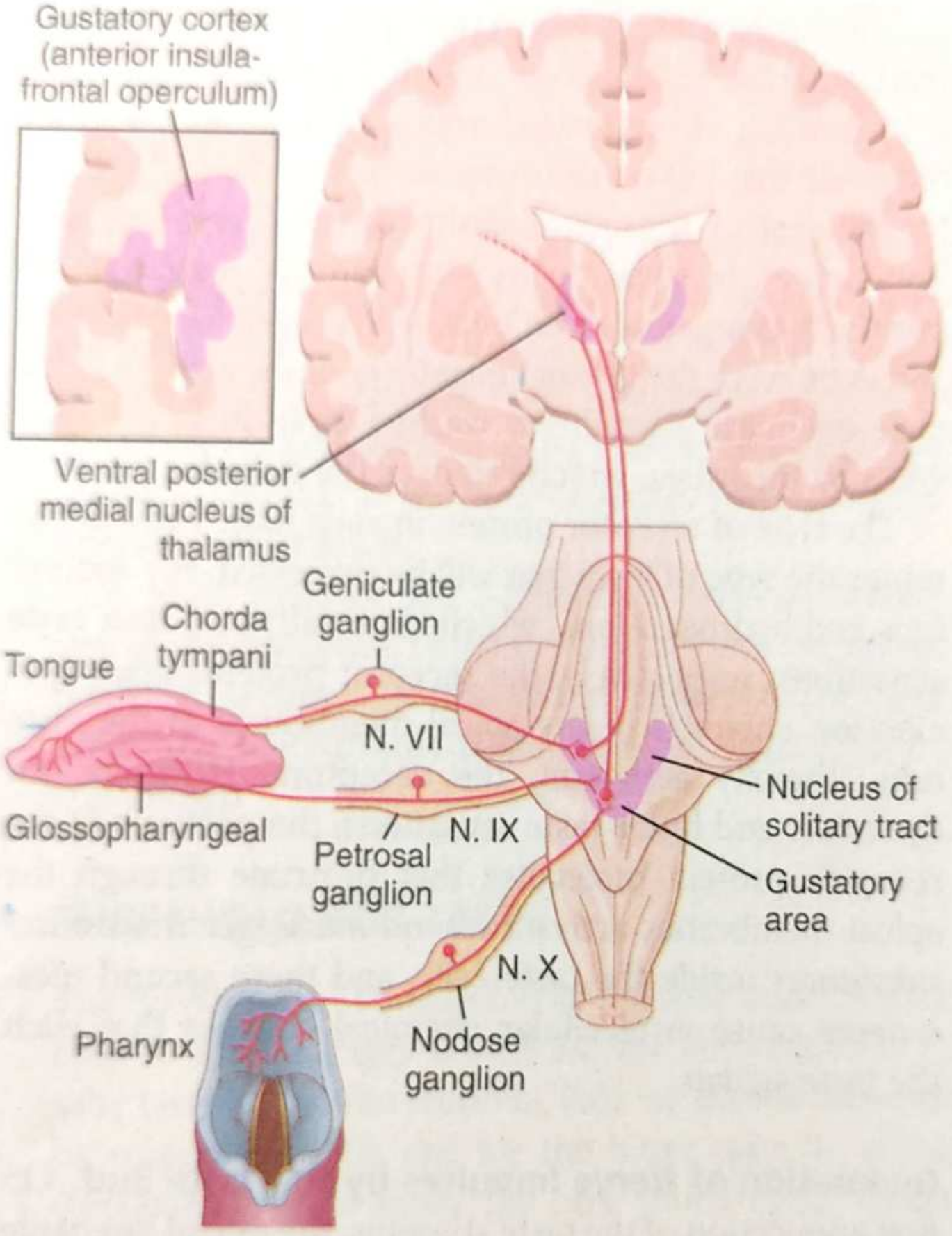
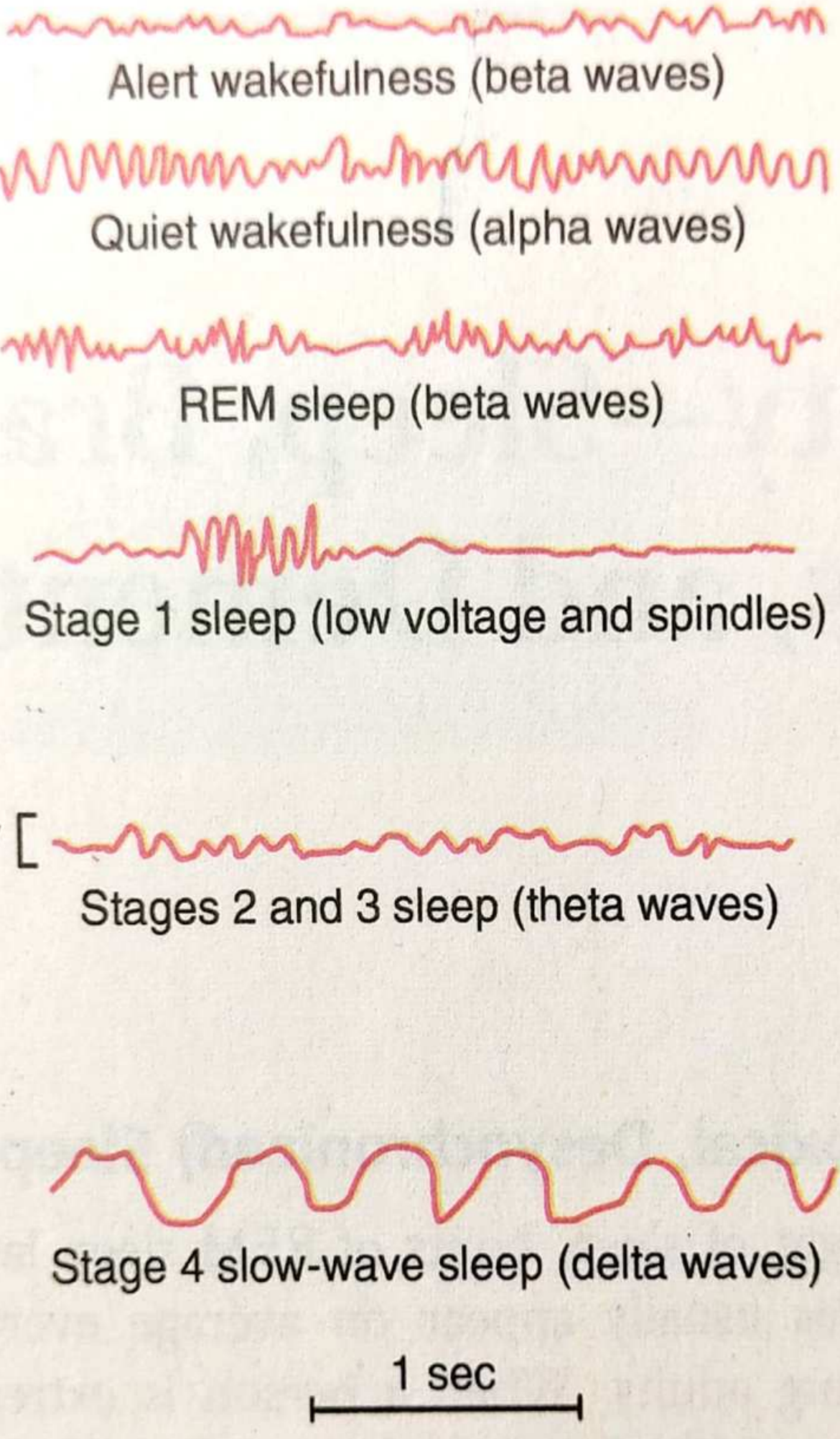


Figure 54-2. Transmission of taste signals into the central nervous system. N., nerve.


Alert wakefulness (beta waves)
Quiet wakefulness (alpha waves)
REM sleep (beta waves)
Stage 1 sleep (low voltage and spindles)
50 μV
Stages 2 and 3 sleep (theta waves)
Stage 4 slow-wave sleep (delta waves)
1 sec

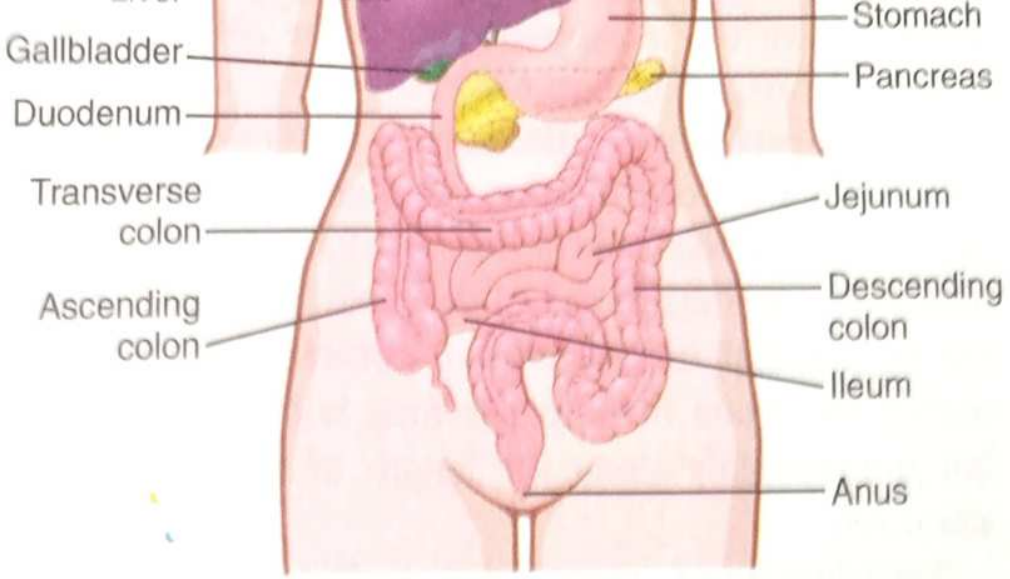


Figure 63-1. Alimentary tract.

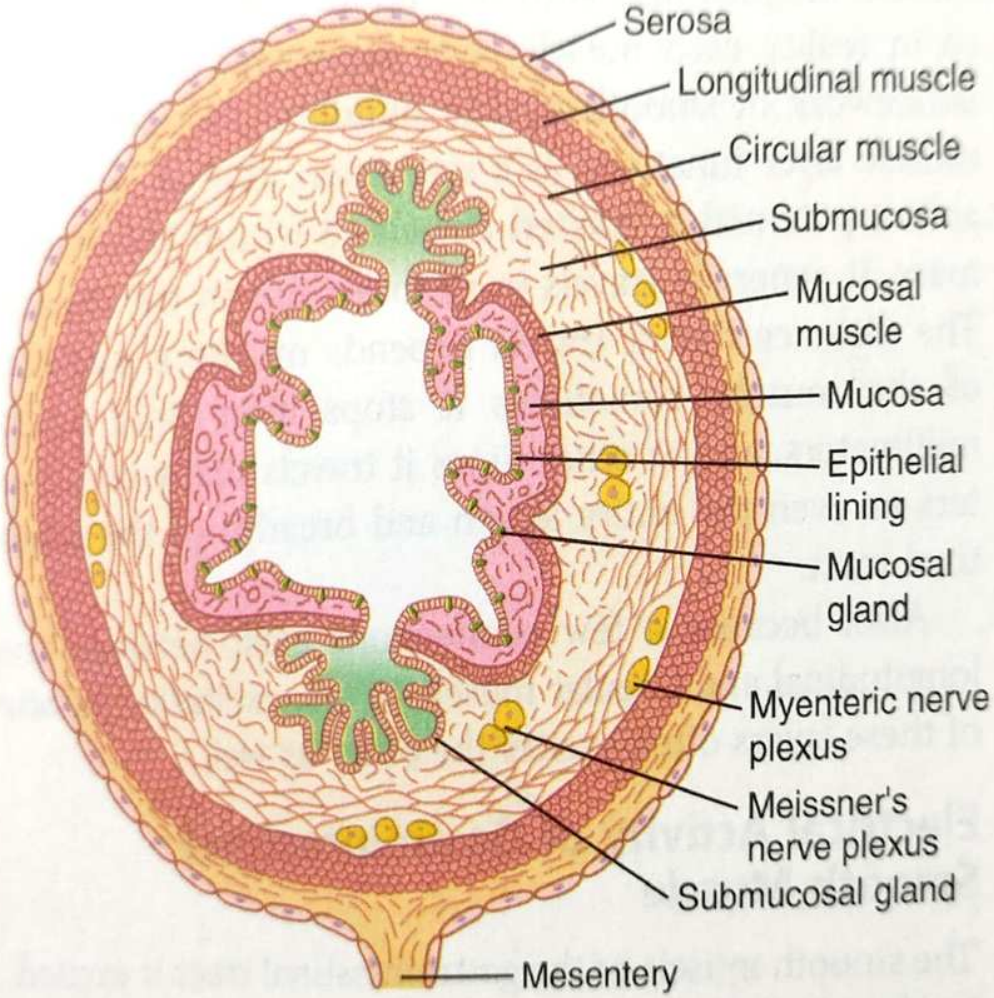


Figure 63-2. Typical cross section of the gut.

are not action potentials. Instead, they are slow, undulating changes in the resting membrane potential. Their intensity usually varies between 5 and 15 millivolts, and their frequency ranges in different parts of the human gastrointestinal tract from 3 to 12 per minute—about 3 in the body of the stomach, as much as 12 in the duodenum, and about 8 or 9 in the terminal ileum. Therefore, the rhythm of contraction of the body of the stomach, the duodenum, and the ileum is usually about 3

tion of growth of the gastric mucosa.

Cholecystokinin (CCK) is secreted by "I" cells in the *mucosa of the duodenum and jejunum* mainly in response to digestive products of fat, fatty acids, and

activity of the stomach and therefore slows emptying of gastric contents into the duodenum when the upper small intestine is already overloaded with food products.

Table 63-1 Gastrointestinal Hormone Actions, Stimuli for Secretion, and Site of Secretion

Hormone	Stimuli for Secretion	Site of Secretion	Actions
Gastrin	Protein Distention Nerve *(Acid inhibits release)*	G cells of the antrum, duodenum, and jejunum	Stimulates Gastric acid secretion Mucosal growth
Cholecystokinin	Protein Fat Acid	I cells of the duodenum, jejunum, and ileum	Stimulates Pancreatic enzyme secretion Pancreatic bicarbonate secretion Gallbladder contraction Growth of exocrine pancreas Inhibits Gastric emptying
Secretin	Acid Fat	S cells of the duodenum, jejunum, and ileum	Stimulates Pepsin secretion Pancreatic bicarbonate secretion Biliary bicarbonate secretion Growth of exocrine pancreas Inhibits Gastric acid secretion
Gastric inhibitory peptide	Protein Fat Carbohydrate	K cells of the duodenum and jejunum	Stimulates Insulin release Inhibits Gastric acid secretion
Motilin	Fat Acid Nerve	M cells of the duodenum and jejunum	Stimulates Gastric motility Intestinal motility

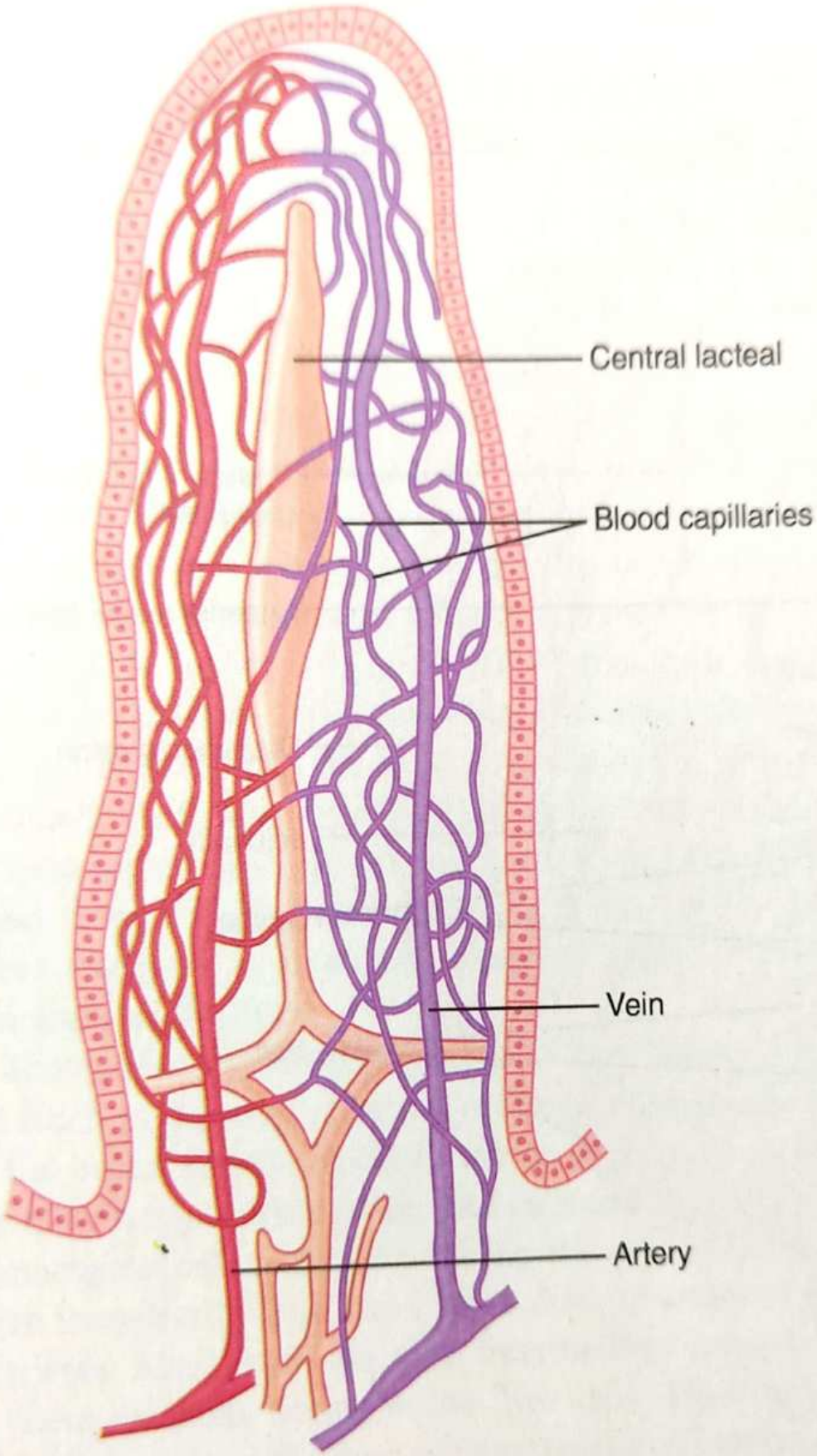


Figure 63-8. Microvasculature of the villus, showing a countercurrent arrangement of blood flow in the arterioles and venules.

the same time that it increases glandular secretion. This increased flow probably results secondarily from the increased glandular activity, not as a direct effect of the nervous stimulation.

Table 65-1 Daily Secretion of Intestinal Juices

Type of Secretion	Daily Volume (ml)	pH
Saliva	1000	6.0-7.0
Gastric secretion	1500	1.0-3.5
Pancreatic secretion	1000	8.0-8.3
Bile	1000	7.8
Small intestine secretion	1800	7.5-8.0
Brunner's gland secretion	200	8.0-8.9
Large intestinal secretion	200	7.5-8.0
Total	6700	

that contains *mucin* for lubricating and for

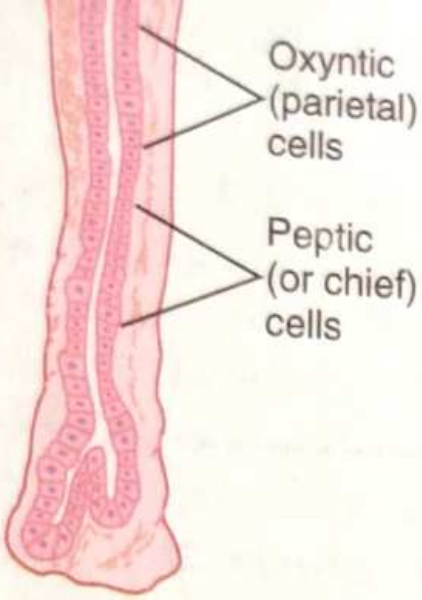


Figure 65-4. Oxyntic gland from the body of the stomach.

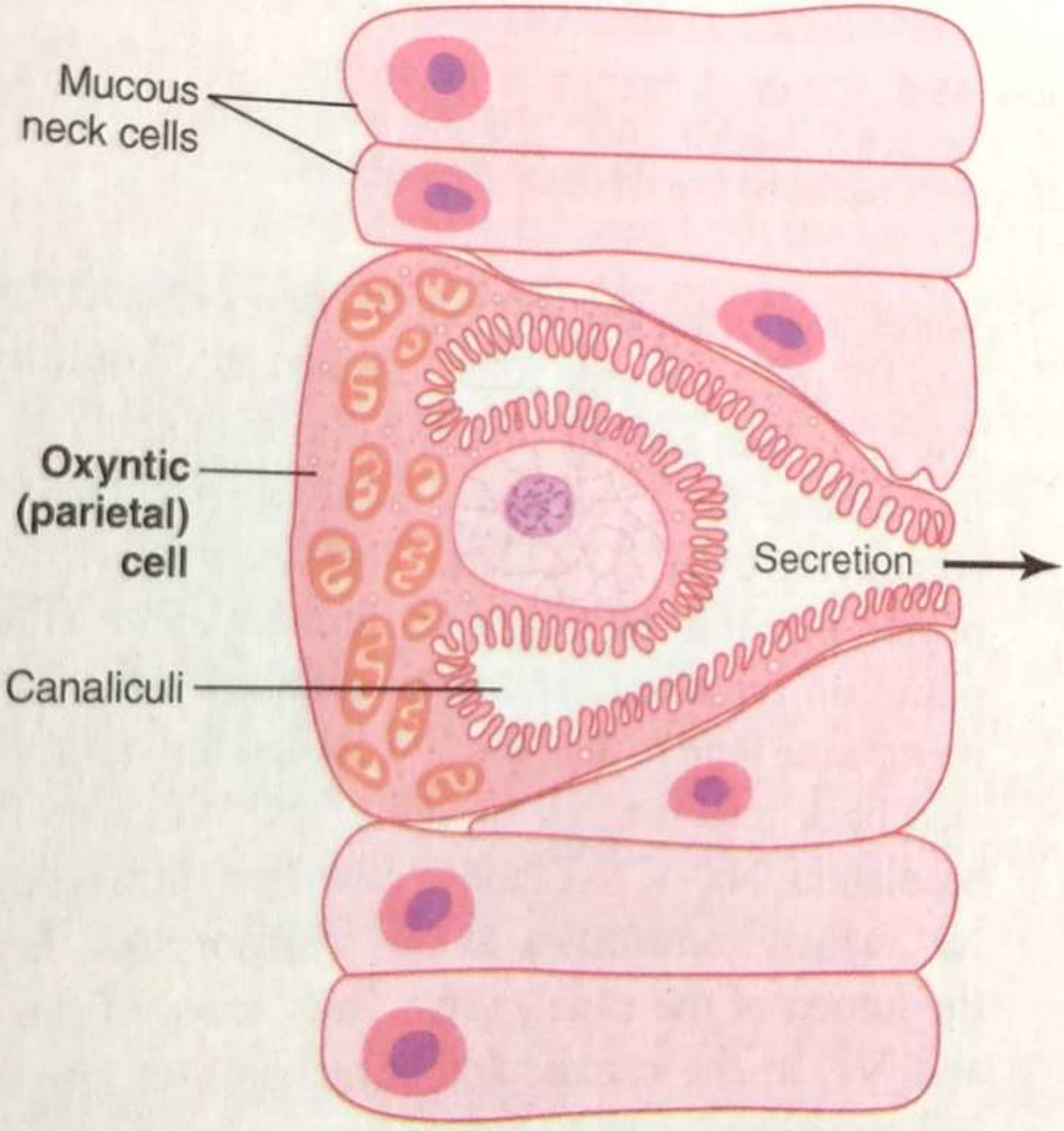


Figure 65-5. Schematic anatomy of the canaliculi in a parietal (oxyntic) cell.

million times that of the arterial blood. To concentrate the

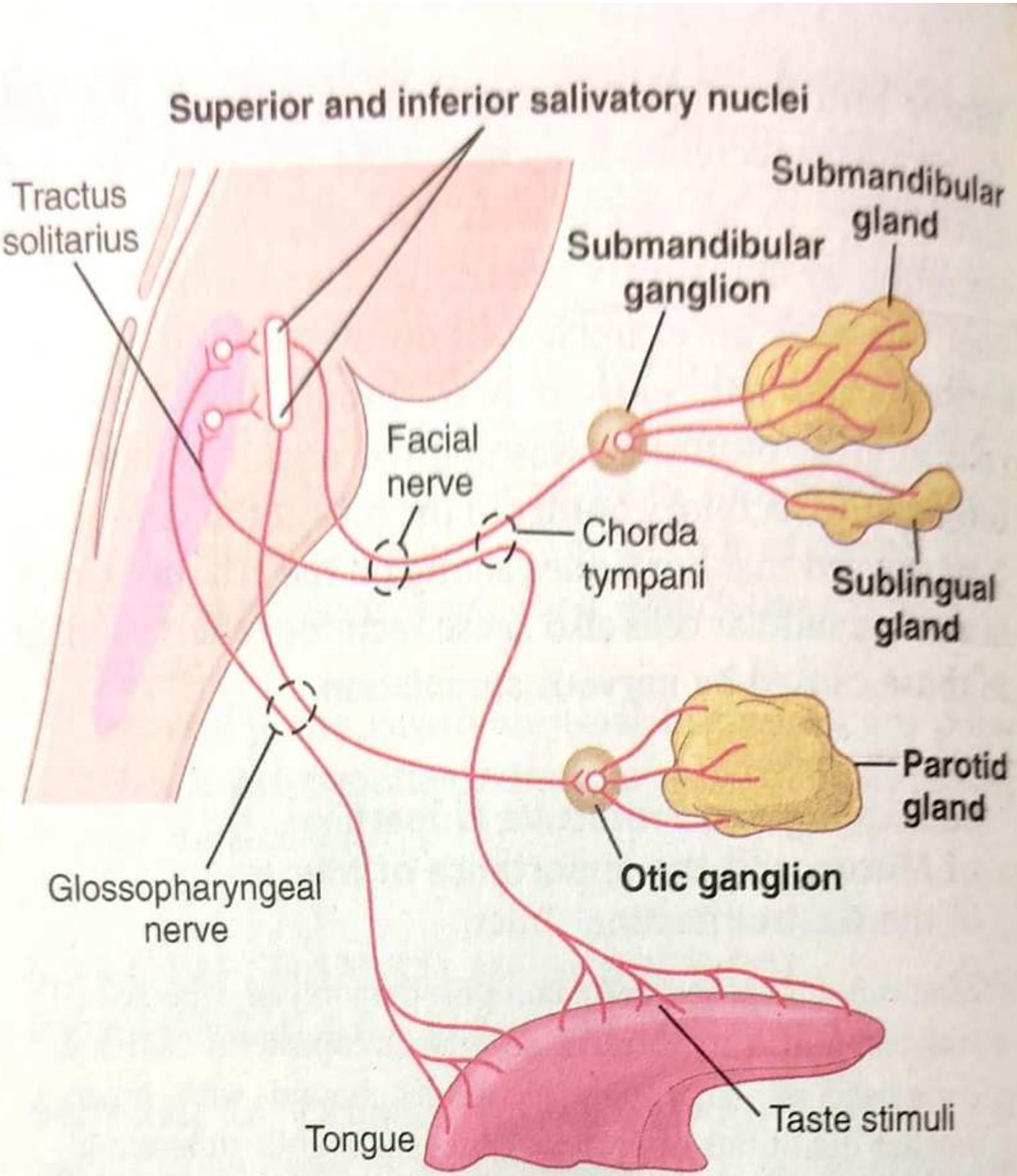


Figure 65-3. Parasympathetic nervous regulation of salivary secretion.

into the duodenum from the stomach).

Figure 65-10. Regulation of pancreatic secretion.

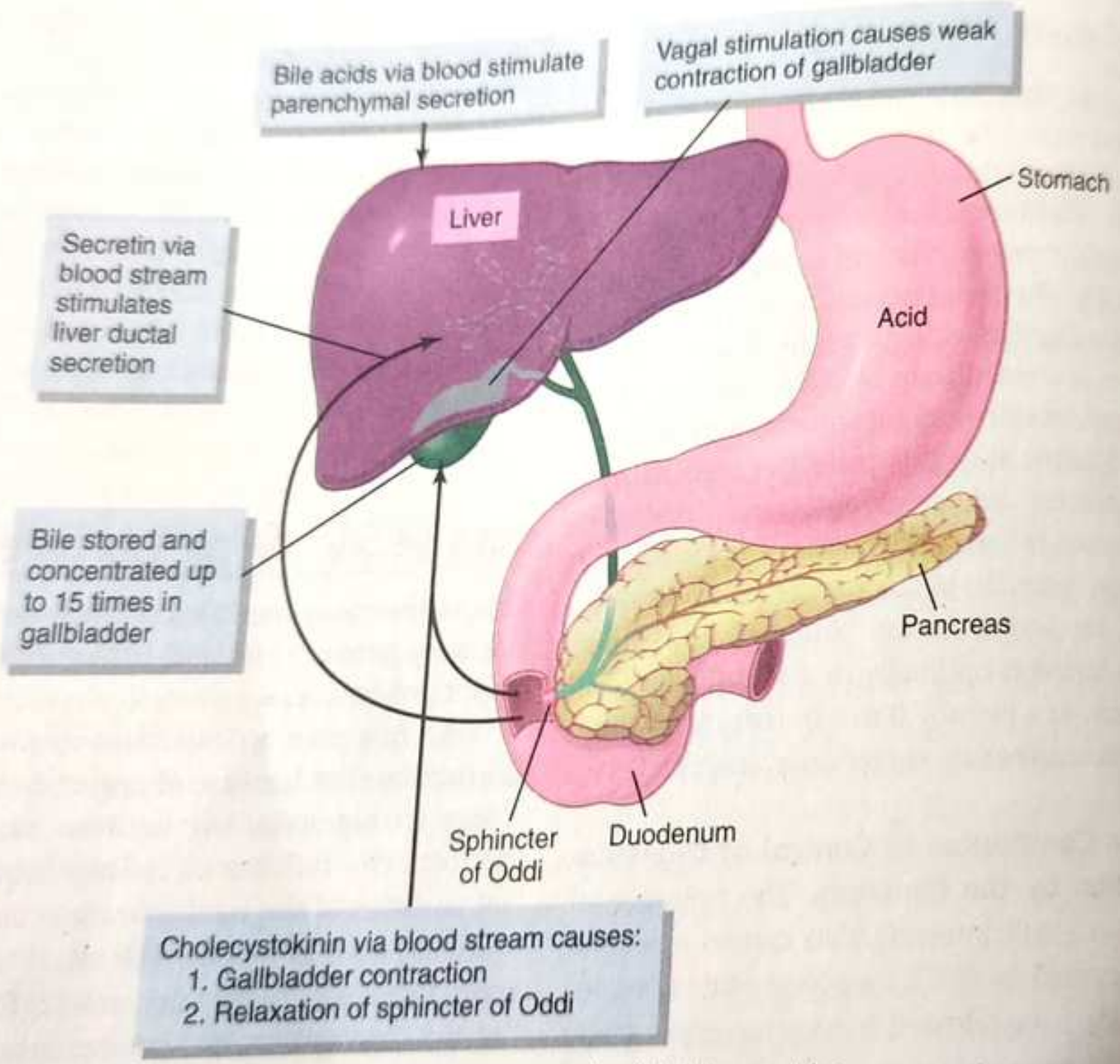


Figure 65-11. Liver secretion and gallbladder emptying.

Table 75-1 Endocrine Glands, Hormones, and Their Functions and Structure

Gland/Tissue	Hormones	Major Functions	Chemical Structure
Hypothalamus (Chapter 76)	Thyrotropin-releasing hormone	Stimulates secretion of thyroid-stimulating hormone and prolactin	Peptide
	Corticotropin-releasing hormone	Causes release of adrenocorticotropic hormone	Peptide
	Growth hormone–releasing hormone	Causes release of growth hormone	Peptide
	Growth hormone inhibitory hormone (somatostatin)	Inhibits release of growth hormone	Peptide
	Gonadotropin-releasing hormone	Causes release of luteinizing hormone and follicle-stimulating hormone	
	Dopamine or prolactin-inhibiting factor	Inhibits release of prolactin	Amine
Anterior pituitary (Chapter 76)	Growth hormone	Stimulates protein synthesis and overall growth of most cells and tissues	Peptide
	Thyroid-stimulating hormone	Stimulates synthesis and secretion of thyroid hormones (thyroxine and triiodothyronine)	Peptide
	Adrenocorticotropic hormone	Stimulates synthesis and secretion of adrenocortical hormones (cortisol, androgens, and aldosterone)	Peptide
	Prolactin	Promotes development of the female breasts and secretion of milk	Peptide
	Follicle-stimulating hormone	Causes growth of follicles in the ovaries and sperm maturation in Sertoli cells of testes	Peptide
	Luteinizing hormone	Stimulates testosterone synthesis in Leydig cells of testes; stimulates ovulation, formation of corpus luteum, and estrogen and progesterone synthesis in ovaries	Peptide
Posterior pituitary (Chapter 76)	Antidiuretic hormone (also called *vasopressin*)	Increases water reabsorption by the kidneys and causes vasoconstriction and increased blood pressure	Peptide
	Oxytocin	Stimulates milk ejection from breasts and uterine contractions	Peptide
Thyroid (Chapter 77)	Thyroxine (T_4) and triiodothyronine (T_3)	Increases the rates of chemical reactions in most cells, thus increasing body metabolic rate	Amine
	Calcitonin	Promotes deposition of calcium in the bones and decreases extracellular fluid calcium ion concentration	Peptide
Adrenal cortex (Chapter 78)	Cortisol	Has multiple metabolic functions for controlling metabolism of proteins, carbohydrates, and fats; also has anti-inflammatory effects	Steroid
	Aldosterone	Increases renal sodium reabsorption, potassium secretion, and hydrogen ion secretion	Steroid
Adrenal medulla (Chapter 61)	Norepinephrine, epinephrine	Same effects as sympathetic stimulation	Amine
Pancreas (Chapter 79)	Insulin (β cells)	Promotes glucose entry in many cells, and in this way controls carbohydrate metabolism	Peptide
	Glucagon (α cells)	Increases synthesis and release of glucose from the liver into the body fluids	Peptide
Parathyroid (Chapter 80)	Parathyroid hormone	Controls serum calcium ion concentration by increasing calcium absorption by the gut and kidneys and releasing calcium from bones	Peptide
Testes (Chapter 81)	Testosterone	Promotes development of male reproductive system and male secondary sexual characteristics	Steroid
Ovaries (Chapter 82)	Estrogens	Promotes growth and development of female reproductive system, female breasts, and female secondary sexual characteristics	Steroid
	Progesterone	Stimulates secretion of "uterine milk" by the uterine endometrial glands and promotes development of secretory apparatus of breasts	Steroid

Continued

Table 75-1 Endocrine Glands, Hormones, and Their Functions and Structure (Continued)

Gland/Tissue	Hormones	Major Functions	Chemical Structure
Placenta (Chapter 83)	Human chorionic gonadotropin	Promotes growth of corpus luteum and secretion of estrogens and progesterone by corpus luteum	Peptide
	Human somatomammotropin	Probably helps promote development of some fetal tissues, as well as the mother's breasts	Peptide
	Estrogens	See actions of estrogens from ovaries	Steroid
	Progesterone	See actions of progesterone from ovaries	Steroid
Kidney (Chapter 26)	Renin	Catalyzes conversion of angiotensinogen to angiotensin I (acts as an enzyme)	Peptide
	1,25-Dihydroxycholecalciferol	Increases intestinal absorption of calcium and bone mineralization	Steroid
	Erythropoietin	Increases erythrocyte production	Peptide
Heart (Chapter 22)	Atrial natriuretic peptide	Increases sodium excretion by kidneys, reduces blood pressure	Peptide
Stomach (Chapter 65)	Gastrin	Stimulates hydrogen chloride secretion by parietal cells	Peptide
Small intestine (Chapter 65)	Secretin	Stimulates pancreatic acinar cells to release bicarbonate and water	Peptide
	Cholecystokinin	Stimulates gallbladder contraction and release of pancreatic enzymes	Peptide
Adipocytes (Chapter 72)	Leptin	Inhibits appetite, stimulates thermogenesis	Peptide

G Protein–Linked Hormone Receptors. Many hormones activate receptors that indirectly regulate the membrane-bound trimeric G protein. Additional details of G protein signaling are discussed in Chapter 46 and shown in **Figure 46-7**.

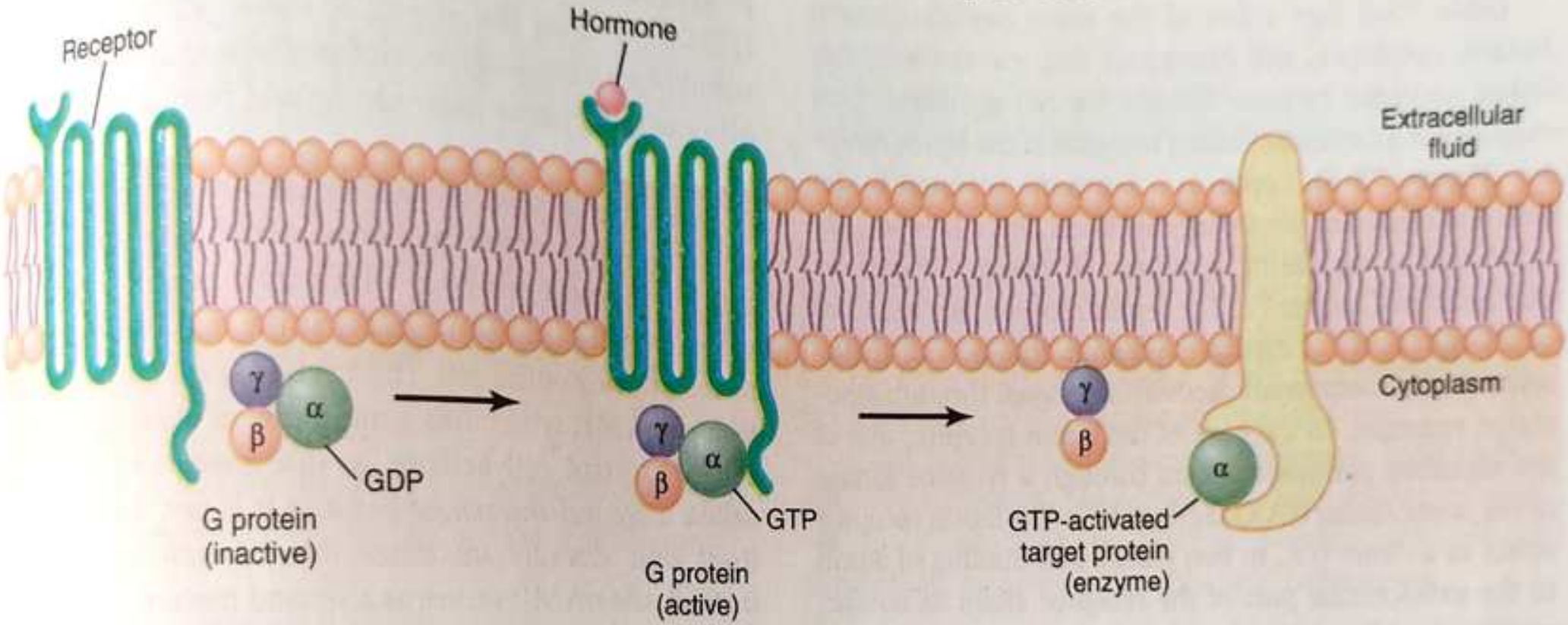


Figure 75-4. Mechanism of activation of a G protein–coupled receptor. When the hormone activates the receptor, the inactive α, β, and γ G protein complex associates with the receptor and is activated, with an exchange of guanosine triphosphate *(GTP)* for guanosine diphosphate *(GDP)*. This process causes the α subunit (to which the GTP is bound) to dissociate from the β and γ subunits of the G protein and to interact with membrane-bound target proteins (enzymes) that initiate intracellular signals.

Table 76-1 Cells and Hormones of the Anterior Pituitary Gland and Their Physiological Functions

Cell	Hormone	Chemistry	Physiological Action
Somatotropes	Growth hormone (GH) (somatotropin)	Single chain of 191 amino acids	Stimulates body growth; stimulates secretion of insulin-like growth factor-1; stimulates lipolysis; inhibits actions of insulin on carbohydrate and lipid metabolism
Corticotropes	Adrenocorticotropic hormone (ACTH) (corticotropin)	Single chain of 39 amino acids	Stimulates production of glucocorticoids and androgens by the adrenal cortex; maintains size of zona fasciculata and zona reticularis of cortex
Thyrotropes	Thyroid-stimulating hormone (TSH) (thyrotropin)	Glycoprotein of two subunits, α (89 amino acids) and β (112 amino acids)	Stimulates production of thyroid hormones by thyroid follicular cells; maintains size of follicular cells
Gonadotropes	Follicle-stimulating hormone (FSH)	Glycoprotein of two subunits, α (89 amino acids) and β (112 amino acids)	Stimulates development of ovarian follicles; regulates spermatogenesis in the testis
	Luteinizing (LH) hormone	Glycoprotein of two subunits, α (89 amino acids) and β (115 amino acids)	Causes ovulation and formation of the corpus luteum in the ovary; stimulates production of estrogen and progesterone by the ovary; stimulates testosterone production by the testis
Lactotropes-Mammotropes	Prolactin (PRL)	Single chain of 198 amino acids	Stimulates milk secretion and production

through another capillary bed in the lower hypothalamus

Table 76-2 Hypothalamic Releasing and Inhibitory Hormones That Control Secretion of the Anterior Pituitary Gland

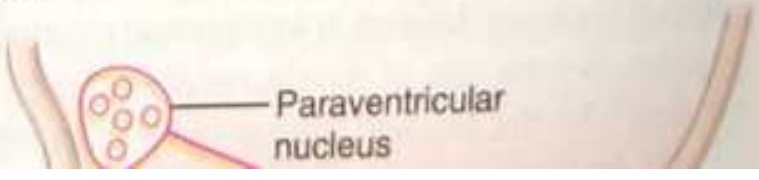
Hormone	Structure	Primary Action on Anterior Pituitary
Thyrotropin-releasing hormone (TRH)	Peptide of 3 amino acids	Stimulates secretion of TSH by thyrotropes
Gonadotropin-releasing hormone (GnRH)	Single chain of 10 amino acids	Stimulates secretion of FSH and LH by gonadotropes
Corticotropin-releasing hormone (CRH)	Single chain of 41 amino acids	Stimulates secretion of ACTH by corticotropes
Growth hormone–releasing hormone (GHRH)	Single chain of 44 amino acids	Stimulates secretion of growth hormone by somatotropes
Growth hormone inhibitory hormone (somatostatin)	Single chain of 14 amino acids	Inhibits secretion of growth hormone by somatotropes
Prolactin-inhibiting hormone (PIH)	Dopamine (a catecholamine)	Inhibits synthesis and secretion of prolactin by lactotropes

ACTH, adrenocorticotropic hormone; FSH, follicle-stimulating hormone; LH, luteinizing hormone; TSH, thyroid-stimulating hormone.

secretion of the anterior pituitary hormones. For most of the anterior pituitary hormones, it is the releasing hormones that are important, but for prolactin, a hypothalamic different hypothalamic releasing or inhibitory hormones are still poorly known, so it would be misleading to attempt delineation here.



Figure 76-8. Patient with acromegaly.





atient with acromegaly.

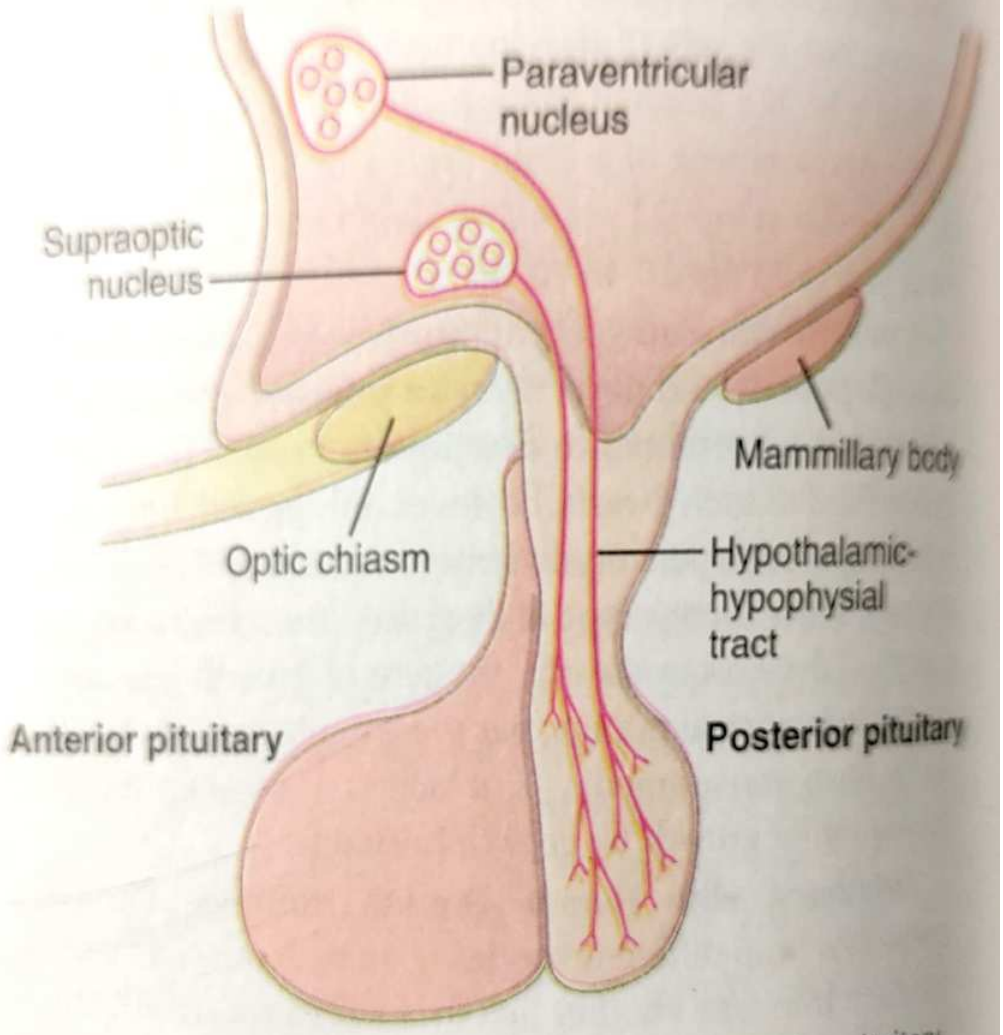


Figure 76-9. Hypothalamic control of the posterior pituitary.

Figure 76-9. These tracts pass to the neurohypophysis through the *pituitary stalk* (hypophysial stalk). The nerve endings are bulbous knobs that contain many secretory granules. These endings lie on the surfaces of capillaries, where they secrete two posterior pituitary hormones: (1) *antidiuretic hormone* (ADH), also called *vasopressin*, and (2) *oxytocin*.

If the pituitary stalk is cut above the pituitary gland but the entire hypothalamus is left intact, the posterior pitu-

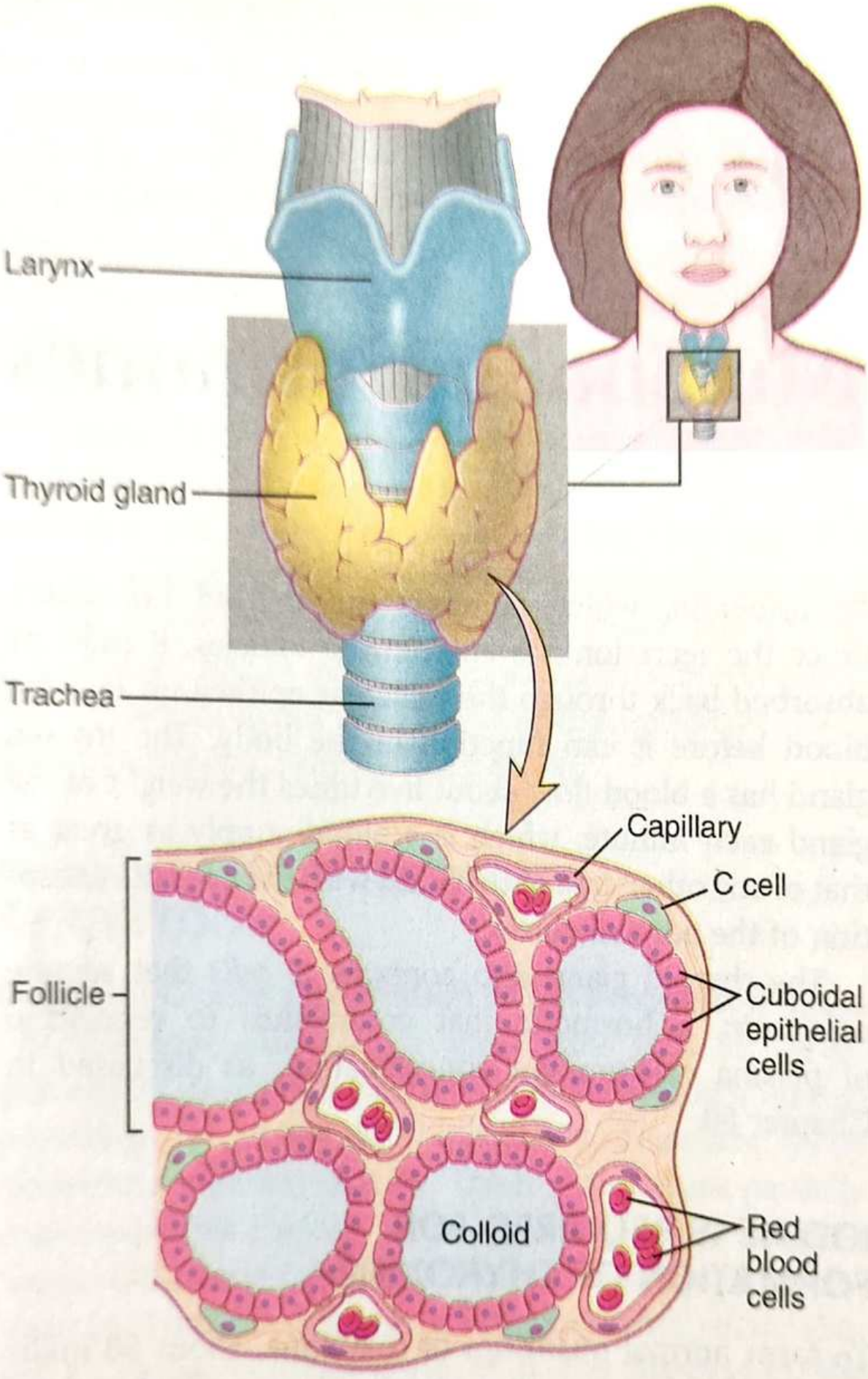


Figure 77-1. Anatomy and microscopic appearance of the thyroid gland, showing secretion of thyroglobulin into the follicles.

causes the total quantity of interstitial fluid to increase.



Figure 77-9. A patient with myxedema. *(Courtesy Dr. Herbert Langford.)*

Cretinism

Cretinism is caused by extreme hypothyroidism during fetal life, infancy, or childhood. This condition is characterized especially by failure of body growth and by mental retardation. It results from congenital lack of a thyroid gland (*congenital cretinism*), from failure of the thyroid gland to produce thyroid hormone because of a genetic defect of the gland, or from a lack of iodine in the diet (*endemic cretinism*).

A neonate without a thyroid gland may have a normal appearance and function because it was supplied with some (but usually not enough) thyroid hormone by the mother while in utero. A few weeks after birth, however, the neonate's movements become sluggish and both physical and mental growth begin to be greatly retarded. Treatment of the neonate with cretinism at any time with adequate iodine or thyroxine usually causes normal return of physical growth, but unless the cretinism is treated within a few weeks after birth, mental growth remains permanently retarded. This state results from retardation



Figure 78-10. A person with Cushing's syndrome before (*left*) and after (*right*) a subtotal adrenalectomy. *(Courtesy Dr. Leonard Posey.)*

IN
UIDS

plasma is mainly in two
e concentration of $HPO_4^=$
concentration of $H_2PO_4^-$
ıe total quantity of phos-
ises, so does the quantity
phosphate ions. Further-
racellular fluid becomes
increase in $H_2PO_4^-$ and
e opposite occurs when
ılkaline. These relations
of acid-base balance in

ine chemically the exact
n the blood, ordinarily
expressed in terms of
liter (100 milliliters) of
inorganic phosphorus



Figure 80-2. Hypocalcemic tetany in the hand, called *carpopedal spasm.*

calcium concentration, and it is usually lethal at about 4 mg/dl.

In laboratory animals, extreme hypocalcemia can

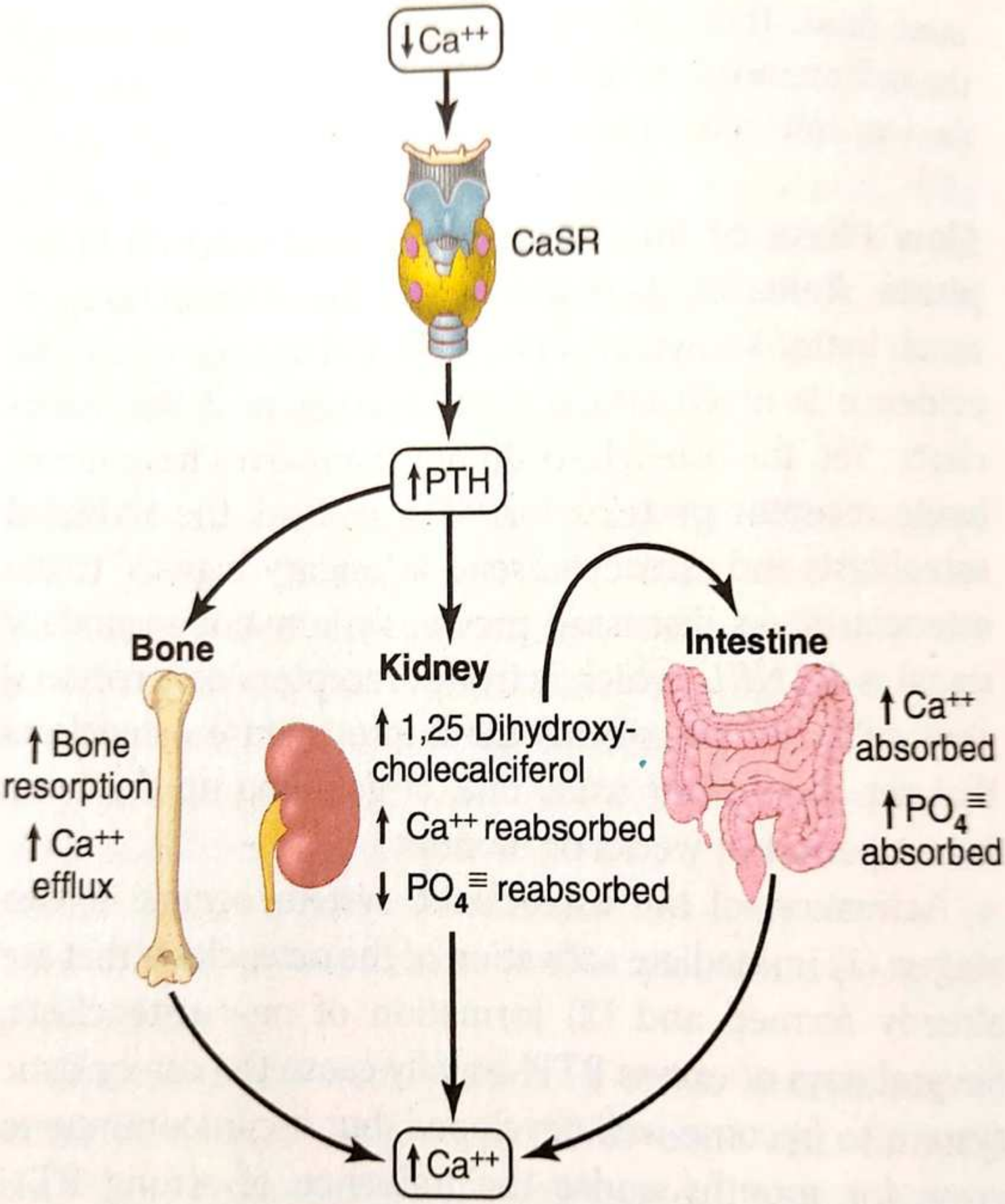


Figure 80-13. Summary of effects of parathyroid hormone *(PTH)* on bone, the kidneys, and the intestine in response to decreased extracellular fluid calcium ion concentration. CaSR, calcium-sensing receptor.

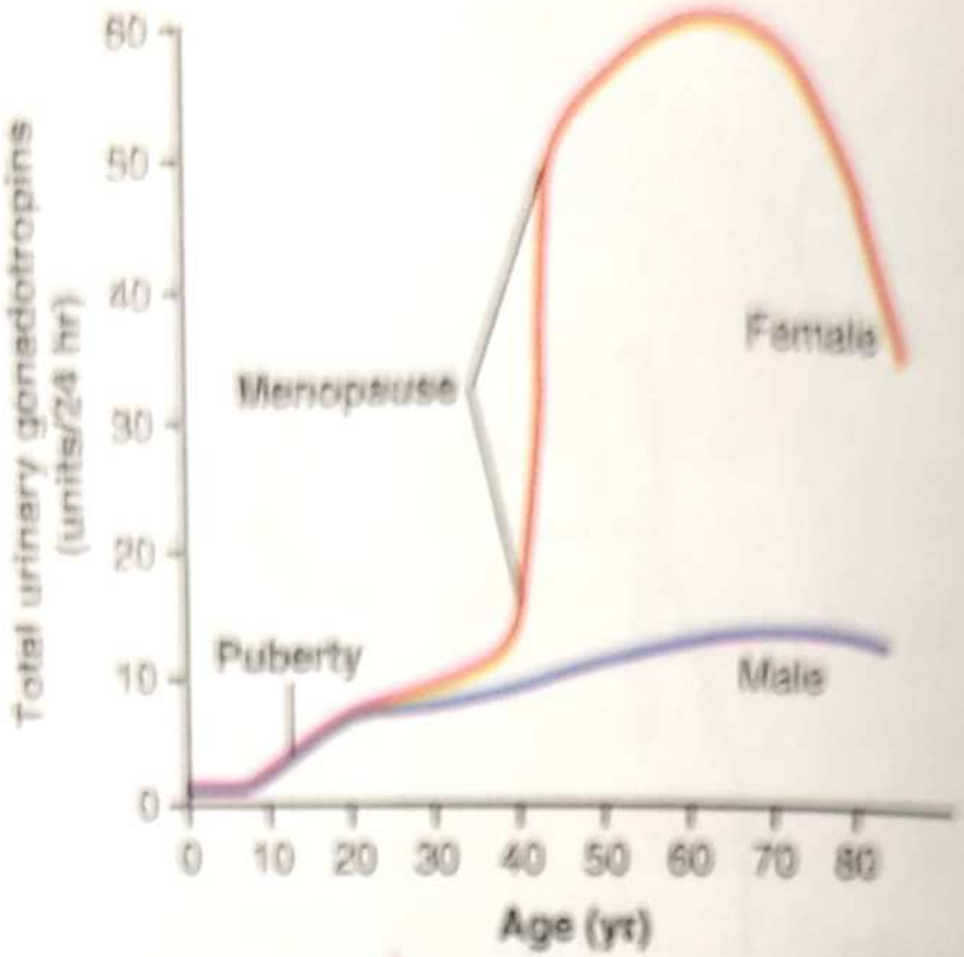


Figure 82-12. Total rates of secretion of gonadotropic hormones throughout the sexual lives of female and male human beings, showing an especially abrupt increase in gonadotropic hormones at menopause in the female.

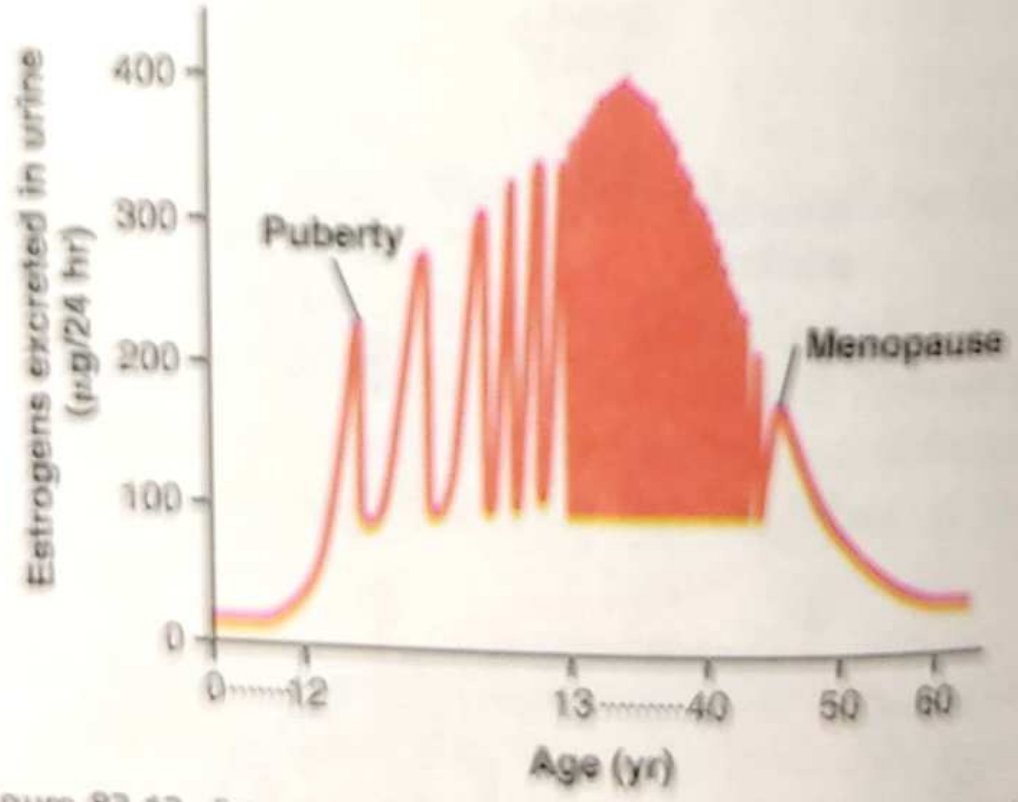


Figure 82-13. Estrogen secretion throughout the sexual life of the female human being.

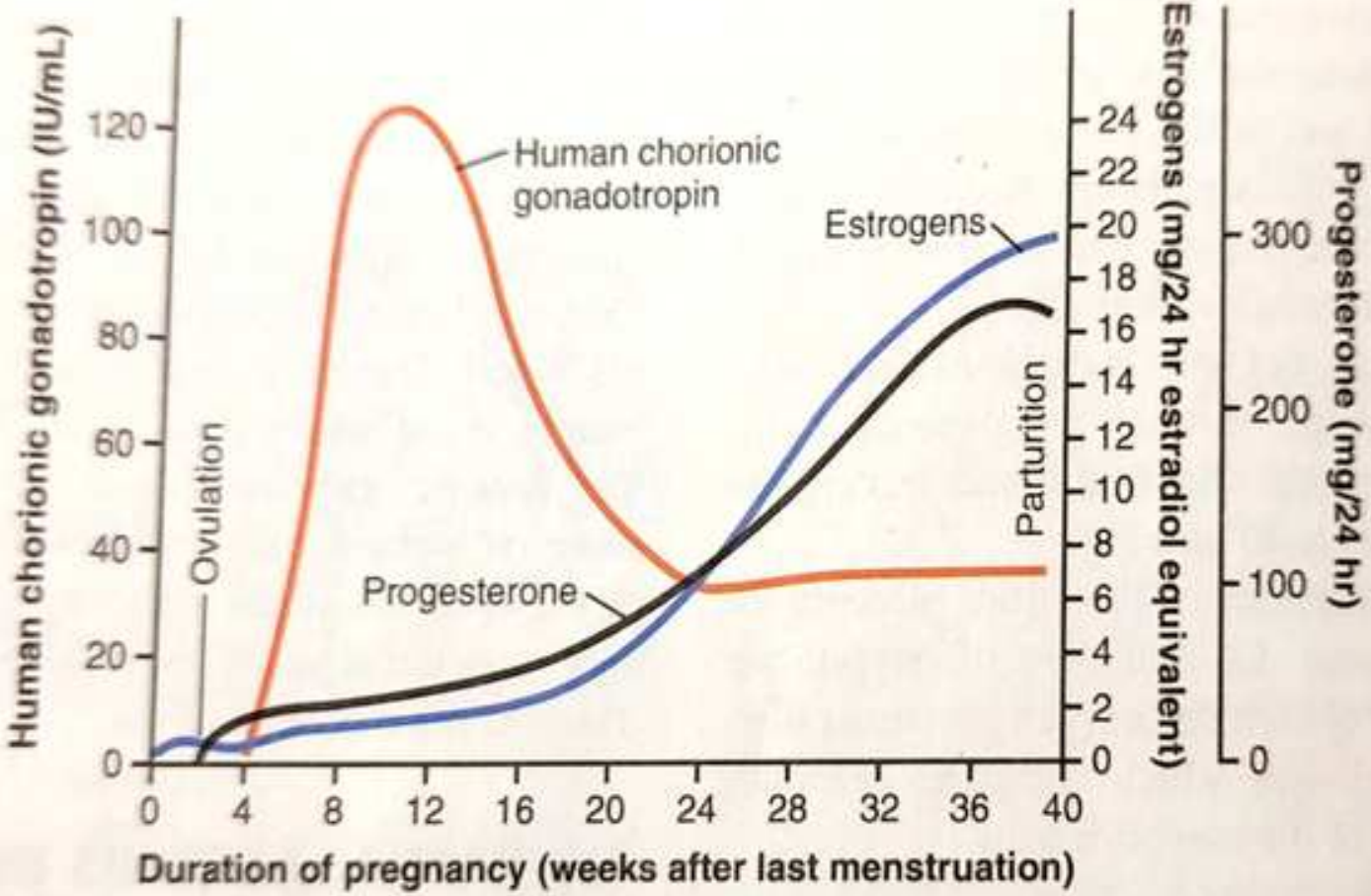
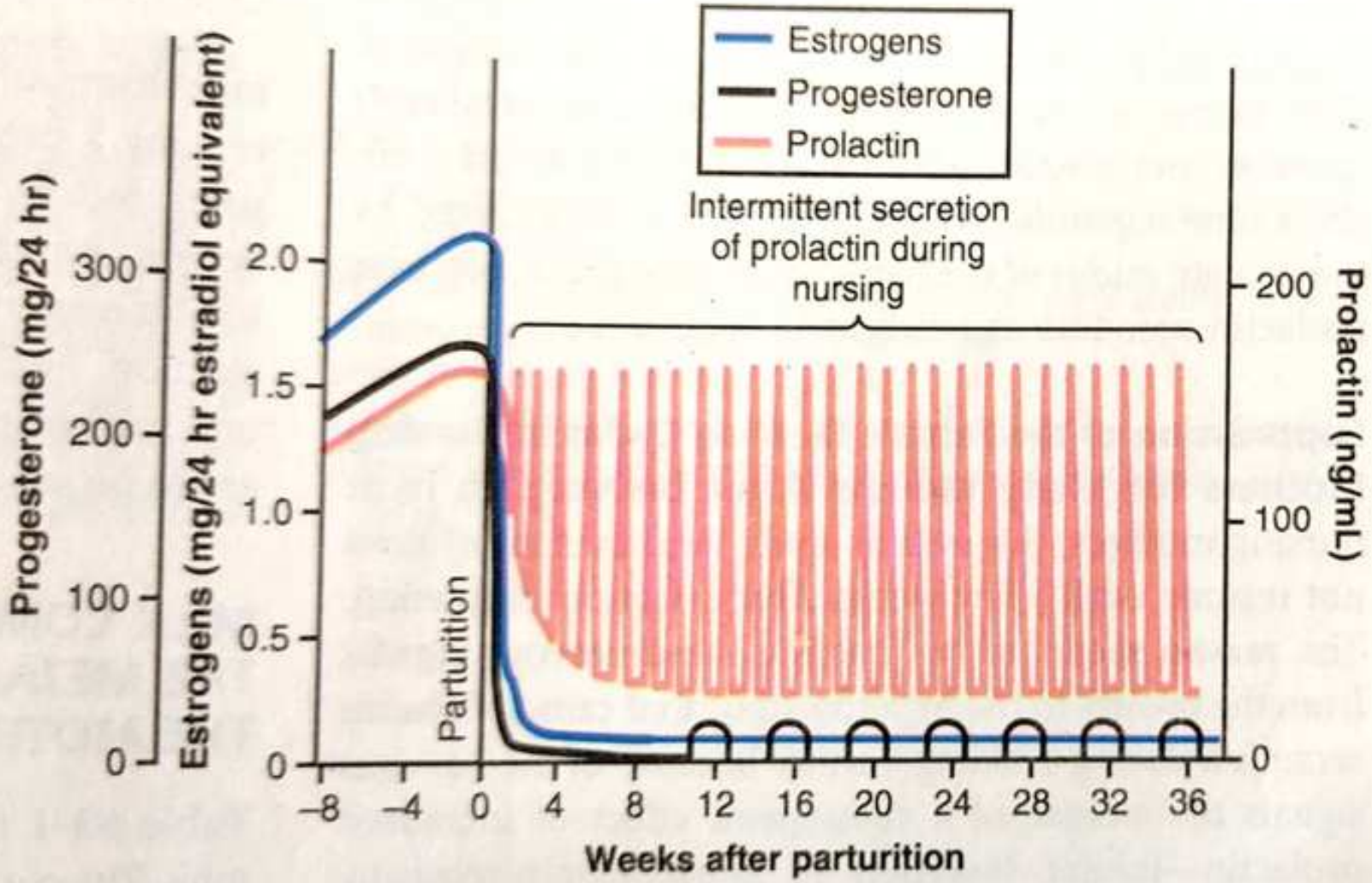


Figure 83-7. Rates of secretion of estrogens and progesterone and concentration of human chorionic gonadotropin at different stages of pregnancy.

cretory lobules, alveoli, and
nstitute its mammary gland
B) and milk-secreting cells of

production.

Immediately after the baby is born, the sudden loss of both estrogen and progesterone secretion from the placenta allows the lactogenic effect of prolactin from the



cretion of
in for 8
; thereaf-
prolactin
w weeks
t periods
t 1 hour
ursing.